Breaking Out of the Adelic Semigroup in Mahler's $3/2$ Problem: Post-Bit-Length Obstructions, Ceiling-Dual Gauge Renormalization, Parry Kneading Dynamics, and a 154-Theorem Lean 4 Library Mapping Seven Structural Barriers

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Cordoned-Off Verification Artifact: `semigroup_breakout/SemigroupBreakoutTheorems.lean` (`154` theorems `SB.1–SB.154`, `13,280+` lines, `0` `sorry`s, `0` warnings, Lean 4.19.0 + Mathlib v4.19.0; companion PDF `SEMIGROUP_BREAKOUT_PAPER.pdf`)
Computational Suite: `loop1_product_formula_and_pade.py`, `loop1_beukers_pade_and_hankel.py`, `loop2_pade_lll_and_divergence.py`, `loop3_archimedean_ceiling_and_small_u.py`, `loop4_flp_sparsity_and_repulsion.py`, `sandbox_symmetry_breaking_loops.py`, `sandbox_wave2_towers.py` through `sandbox_wave7_binary_carry_bridge.py`, `sandbox_overnight_20_bridge_loops.py`, and `sandbox_wave8_jk_diagonal_asymptotics.py` through `sandbox_wave27_capstone_rank_compression_and_barrier7.py`

Abstract

Let $Z := \{\alpha \in \mathbb{R}_{>0} : \forall n \in \mathbb{N},\ \{\alpha(3/2)^n\} \in [0, 1/2)\}$ denote Mahler's set of Z-numbers, conjectured since Mahler (1968) to be empty ($Z = \emptyset$). Building on our 226-file bottom-up Lean 4 campaign (`MAHLER_FORMALIZATION_PAPER.pdf`), we examine whether formal-kernel-guided proof search merely interpolates inside the *convex hull* of known theorems or genuinely explores the *semigroup* generated by elementary algebraic, cocycle, and modular rules. We first formulate an **Adelic Finite-Horizon Decoupling Principle** (Section 2 and `SB.109`) explaining why purely local congruence and cocycle generators hit a structural wall: at any finite horizon K, the integer state equations $2^{T_r} P_r = 3^{T_r} P_0 + A_{0,r}(s)$ and real companion equations $2^{T_r} h_r = 3^{T_r} h_0 - A_{0,r}(s)$ decouple P_0 from h_0 and admit simultaneous solutions modulo 2^{T_K+1} and modulo every odd integer $M = 2m + 1$ (`SB.109`), so every Parry-admissible K-block word is realized by a positive odd integer $b_0 \in (0, 2^{T_K+1}) \subset \mathbb{N}$.

Guided by the resulting **Post-Bit-Length Exhaustion Principle** (symmetry-breaking generators must invoke archimedean positivity $u_K \geq 1$ and seed finiteness $P_0 < 2^{T_K}$), we formalize **154 `sorry`-free, warning-free theorems (`SB.1–SB.154`)** in `SemigroupBreakoutTheorems.lean`. Principal contributions include: 1. **Machine-Checked Real Z-Number $(3/2)^W$ Bound, Parry Kneading Renewal, & Exact $(N+1)$ Width Compression (`SB.111–SB.112`, `SB.121`, `SB.141`, `SB.149`, `SB.153`):** Formalizes Flatto's (1992) and Flatto–Lagarias–Pollington's (1995) effective bound $\#(Z \cap [0, X)) \leq |\mathcal{C}_W| \leq 3(3/2)^W - 2$ ($X \leq 2^W$, giving $O(X^{\log_2(3/2)}) \approx O(X^{0.58496})$) directly over $\mathbb{R}$ (`SB.153`, including Mahler's uniqueness lemma per unit interval $[b_0, b_0 + 1)$) via exact 3^W scaled-width conservation (`SB.112`, `SB.121`), establishes the exact Parry kneading spine renewal decomposition of $|\mathcal{C}_N|$ (`SB.141`, enabling $O(N^2)$ dynamic-programming evaluation of individual 2-adic Fourier modes out to $N = 300$), and proves that the set of distinct scaled cylinder widths at depth N has exact cardinality $N + 1$ (`SB.149`). 2. **The Ceiling-Dual Gauge $W_r = (5P_r + 4)/36$, Super-Block Renormalization, & Real-Companion $159/160$ Mod-320 Pinch (`SB.81–SB.88`, `SB.97–SB.98`, `SB.115`, `SB.138–SB.139`, `SB.154`):** Introduces a linear gauge that annihilates additive error on $s_r = 2$ steps ($4W_{r+1} = 9W_r$), reduces consecutive Super-Blocks at $s^* = 3$ to the Catalan–Pillai unit equation $3^{2m_1+3}V_1 = 2^{2m_2+3}V_2 + 1$ with $\gcd(V_1, V_2) = 1$ (`SB.115`), rules out short post-2 recoveries $c \in \{1, 2, 3, 4, 5\}$ on $[0, 1/2)$ (`SB.138`), and proves directly over $\mathbb{R}$ (`SB.154`) that every surviving real companion with $s_r \geq 2$ and $s_{r+1} = 2$ forces $s_{r+2} \geq 6$ and $W_{r+1} \equiv 4 \pmod{320}$ (eliminating 159 of 160, or 99.375%, of all even residue classes modulo 320). 3. **Kempner Threshold Inversion Across Odd Fractions $q/2$ (`SB.105–SB.108`):** Proves that while every odd fraction $q/2$ ($q \geq 3$) shares the one-step mod-4 50/50 split and the $(q^2 - 4)u + 1$ core gauge, $q = 3$ $(3/2)$ is the unique odd fraction whose Zone-1 canyon suffers an Archimedean Kempner overflow ($8/5 = 1.60 > 1$). 4. **Seven Structural Barriers & Conditional Sub-FLP Reductions (`SB.109`, `SB.116`, `SB.123–SB.135`, `SB.143–SB.148`, `SB.150`):** Combines formal obstructions in Lean with classical Diophantine and analytic thresholds (Baker–Matveev, Ridout Q^{-2}, Schlickewei Subspace, Pólya–Carlson $R_\infty R_2 = 1$, Parseval odd-mode floors, and standalone abc / shift-average / rogue-reset limits) to map why existing approaches stall short of $Z = \emptyset$ (which remains open), alongside explicit conditional reductions to $O(X^{5/12}) \approx O(X^{0.4167})$ (`SB.123`) and $O(X^{\log_2(15)/7}) \approx O(X^{0.55813})$ (`SB.144`).

1. Introduction: Convex Hull vs. Semigroup & Scope of Formalization

A central question in mathematical formalization and automated reasoning is characterizing the reach of compositional proof search: - **The Convex-Hull Hypothesis:** Automated search only interpolates between existing proofs in the literature. - **The Semigroup Hypothesis:** When coupled with symbolic computation and a formal proof kernel (Lean 4), search can compose elementary proof generators $g_1, g_2, \ldots, g_k$ into long chains $g_{i_1} \circ g_{i_2} \circ \cdots \circ g_{i_m}$, reaching true, non-trivial theorems outside the convex hull of prior papers, provided a path exists within the semigroup generated by $\{g_1, \ldots, g_k\}$.

Our 226-file bottom-up formalization of Mahler's $3/2$ problem demonstrated both the reach and the boundary of this semigroup regime: - **Outside the Convex Hull:** Chaining computational probing with Lean verification across 30 outer loops produced a `sorry`-free theory of the sub-ceiling corridor ($0 < \Sigma_N < 2^{T_N}$), proving that it contains a unique, non-branching, strictly aperiodic infinite sequence $\kappa : \mathbb{N} \to \mathbb{N}$ governed by $\widehat{\Sigma}_{N+1}(m+1) = 3\Sigma_{N+1}(m)$ and $\Sigma_{N+M}(s) = 3^d \widehat{\Sigma}_N(s) + 2^{T_N}\Sigma_M(\mathrm{shift}_N s)$. - **Inside a Closed Sub-Semigroup:** Every elementary generator in those loops (ring identities, linear integer inequalities, cocycle composition, and mod-2^k/mod-3^k congruences) holds not only in $\mathbb{N}$ but also in $\mathbb{Z}_2$ (and across $\widehat{\mathbb{Z}}$). Because the composition of $\mathbb{Z}_2$-valid rules remains $\mathbb{Z}_2$-valid, purely local congruence/cocycle chains cannot leave the $\mathbb{Z}_2$-compatible universe where Mahler's conjecture fails by Hensel's lemma.

To leave that sub-semigroup, new generators must invoke **archimedean positivity and finiteness** ($1 \le u_K$ and $P_0 < 2^W$).

Scope of Formalization

To maintain strict transparency between machine-checked statements and surrounding mathematical interpretation: - **Machine-Checked in Lean 4 (`SB.1–SB.154`):** All 154 theorems in `SemigroupBreakoutTheorems.lean` compile under Lean 4.19.0 (`-DwarningAsError=true`) using only Lean's standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), with zero `sorry`, `admit`, custom `axiom`, or `native_decide`. Theorems `SB.1–SB.152` are formulated over $\mathbb{N}$ and $\mathbb{Z}$, while `SB.153` and `SB.154` bridge directly to $\mathbb{R}$ via `ValidCompanionUpTo`, `ValidCompanionForever`, and `Int.fract` from the root library. - **Analytic Interpretations & Reproducible Computation (Informal):** Complex-valued character sums $\mathcal{S}(h;N) \in \mathbb{C}$, Plancherel/Walsh–Hadamard unitary steps interpreting `SB.122` / `SB.124` / `SB.127`, the continuation-operator rank interpretation of `SB.149`'s exact $(N+1)$-width theorem, the $O(N^2)$ dynamic-programming algorithm for $\mathcal{S}(h;N)$ (`sandbox_wave26_twisted_operator_and_laplace_partition.py`), and external transcendence/analytic citations (Matveev 2000, Ridout 1957, Schlickewei 1990, Pólya 1921) are presented analytically in the text rather than formalized over $\mathbb{C}$.

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2. The Finite-Horizon Decoupling Meta-Theorem & Post-Bit-Length Free-Fall

2.1 Why No Local K-Block Auxiliary Polynomial Can Have Product-Formula Ratio $\rho(F) < 1$

In classical diophantine approximation (Thue, Siegel, Roth, Baker, Beukers), one constructs a non-zero integer $Z = F(P_0, \dots, P_K) \in \mathbb{Z} \setminus \{0\}$ that is divisible by a large power $2^a 3^b$ while bounded in real magnitude by $|Z|_\infty < 2^a 3^b$, violating the product formula $|Z|_\infty \cdot |Z|_2 \cdot |Z|_3 \ge 1$.

Why does every such local K-block auxiliary polynomial $F(P_0, \dots, P_K)$ fail to reach $\rho(F) = |Z|_\infty / (2^a 3^b) < 1$ when constrained only by K-block Mahler survival?

Meta-Theorem 2.1 (Finite-Horizon Decoupling & Realizability Barrier).
At any fixed finite horizon $K \in \mathbb{N}$ and for any block sequence $s = (s_0, \dots, s_{K-1})$ with $s_r \ge 1$: 1. Every integer boundary state $P_r = 3b_r + 1$ ($1 \le r \le K$) is an affine-linear function of P_0 alone:

$$2^{T_r} P_r = 3^{T_r} P_0 + A_{0,r}(s).$$

2. Every real companion tail $h_r = 3\alpha(3/2)^{T_r} - P_r$ ($0 \le r \le K$) is an affine-linear function of h_0 alone:

$$2^{T_r} h_r = 3^{T_r} h_0 - A_{0,r}(s).$$

*3. The variables $P_0 \in \mathbb{N}$ and $h_0 \in \mathbb{R}$ are **completely decoupled** at horizon K: the archimedean survival conditions $0 < h_r < 1/2$ ($0 \le r \le K$) constrain only $h_0 \in \left(\frac{A_{0,K}(s)}{3^{T_K}}, \frac{A_{0,K}(s)+2^{T_K-1}}{3^{T_K}}\right)$ (which is non-empty iff $\Sigma_r(s) > 0$ for all $1 \le r \le K$), while the integer block chain constrains only $P_0 \bmod 2^{T_K+1}$. 4. Consequently, for every Parry-admissible K-block word s, there exists a genuine positive odd integer $b_0 \in (0, 2^{T_K+1}) \subset \mathbb{N}$ and a genuine real number $\alpha = b_0 + \frac{1+h_0}{3} \in \mathbb{R}$ that survive all K blocks. Because $(P_0, \dots, P_K) \in \mathbb{N}^{K+1}$ is realized by an actual tuple of positive integers, no integer polynomial $Z = F(P_0, \dots, P_K)$ can violate $|Z|_\infty \ge 2^a 3^b$ at horizon K without using an additional hypothesis that excludes $b_0 \in (2^W, 2^{T_K+1})$—namely, the **post-bit-length hypothesis** $P_0 < 2^{T_K}$.*

2.2 Empirical Verification: Post-Horizon Free-Fall of Minimal Engineered Seeds

In Loop 1A (`loop1_product_formula_and_pade.py`), we extracted the unique minimal positive odd seed $b_0^{(N)} \in (0, 2^{T_N+1})$ realizing the first N blocks of the canonical sub-ceiling kneading sequence κ for $N = 1, \dots, 25$ (up to $T_{25} = 123$ binary steps, where $b_0^{(25)}$ is a 124-bit integer) and measured how many *additional* steps $b_0^{(N)}$ survives once $T > T_N$ (where $b_0^{(N)}$ has no engineered bits left):

Horizon N	Engineered Steps T_N	$\mathrm{bitlen}(b_0^{(N)})$	Death Block	Death Step	Extra Steps Past T_N	Death Mechanism
1	2	1	2	3	+1	$s_1 = 1$ (odd-to-odd death)
2	8	4	3	9	+1	$s_2 = 1$ (odd-to-odd death)
5	16	16	6	19	+3	$\Sigma_6 \le 0$ (Parry pinch at $s_5 = 3 < 10$)
6	26	27	7	28	+2	$\Sigma_7 \le 0$ (Parry pinch at $s_6 = 2 < 7$)
10	48	48	11	50	+2	$\Sigma_{11} \le 0$ (Parry pinch at $s_{10} = 2 < 3$)
11	51	51	15	63	+12	$s_{14} = 1$ (after ceiling jump $s_{11} > 2$)
13	65	66	14	67	+2	$\Sigma_{14} \le 0$ (Parry pinch at $s_{13} = 2 < 4$)
15	72	73	16	77	+5	$\Sigma_{16} \le 0$ (Parry pinch at $s_{15} = 5 < 8$)
20	101	102	21	102	+1	$s_{20} = 1$ (odd-to-odd death)
25	123	124	26	125	+2	$\Sigma_{26} \le 0$ (Parry pinch at $s_{25} = 2 < 8$)

- **Result:** Across all 25 horizons, the minimal T_N-bit seed survives an average of only 2.48 **extra binary steps** past T_N (and in 23 **out of** 25 **horizons**, it dies on the **very first un-engineered block** s_N).
- Similarly, our exhaustive census of $b_0 \in [1, 20{,}000]$ (`loop3_archimedean_ceiling_and_small_u.py`) showed that only 8 odd seeds survive to step 25 (`149, 4045, 5325, 8101, 10901, 11573, 12109, 19797`), only 1 (`8101`) survives to step 34, and **all** $20{,}000$ **seeds are extinct by step** 39.

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3. Five Formalized $\mathbb{Z}_2$-Symmetry-Breaking & Breakout Theorems

In `SemigroupBreakoutTheorems.lean`, we formalized five theorems that explicitly break the $\mathbb{Z}_2$ / single-branch symmetry of the bottom-up campaign:

3.1 Theorem SB.1: The Global Archimedean Rung Ceiling (`archimedean_rung_ceiling_and_stepSum_bound`)

In $\mathbb{Z}_2$, a 2-adic integer $b_0 \in \mathbb{Z}_2$ can have arbitrarily large 2-adic valuation $s_K = v_2(3b_K + 1)$ at any step K (for example, $b_0 = (2^{1000} - 1)/3 \in \mathbb{Z}_2$ has $s_0 = 1000$). In $\mathbb{N}$, however, the odd quotient $u_K = P_K / 2^{s_K}$ is a **positive** odd integer ($u_K \geq 1$), which couples with the renewal master equation $2^{T_K} P_K = 3^{T_K} P_0 + A_{0,K}(s)$ and the survival condition $2A_{0,K}(s) < 3^{T_K}$ to yield:

$$2 \cdot 2^{T_{K+1}} \leq 2 \cdot 2^{T_{K+1}} u_K = 2 \cdot 3^{T_K} P_0 + 2A_{0,K}(s) < 3^{T_K}\big(2P_0 + 1\big).$$

Unlike the sub-ceiling bound $s_K \leq 2T_K + 2$ (which applies only while $\widehat{\Sigma}_{K+1} < 0$), Theorem SB.1 bounds s_K and $T_{K+1} = T_K + s_K$ across **all** surviving blocks—even those that restore the full ceiling—and at $K = 0$ gives $2^{s_0} \leq 3b_0 + 1$.

3.2 Theorem SB.2: The Pure Power-of-3 ($u_K = 1$) Collapse & Mod-64 Trap (`unit_quotient_even_rung_instant_death_and_odd_rung_mod64_identity`)

The archimedean ceiling $2^{s_K} \leq P_K$ is sharpest when $u_K = 1$ ($P_K = 2^{s_K}$), which forces the next odd state to be a pure power of 3: $b_{K+1} = 3^{s_K - 1}$. We proved: 1. **Even Rung ($s_K = 2j, j \geq 1$):** $b_{K+1} = 3^{2j-1}$ has $P_{K+1} = 3^{2j} + 1 = 9^j + 1 \equiv 2 \pmod 8$, so $v_2(P_{K+1}) = 1$ (**instant odd-to-odd death on the very next block**). 2. **Odd Rung ($s_K = 2j + 1, j \geq 1$):** $b_{K+1} = 3^{2j}$ has $P_{K+1} = 3^{2j+1} + 1 = 3 \cdot 9^j + 1 \equiv 4 \pmod 8$, forcing a short block $s_{K+1} = 2$ with $u_{K+1} = (3^{2j+1} + 1)/4$ and $b_{K+2} = 3u_{K+1}$. Remarkably, $P_{K+2} = 3b_{K+2} + 1$ obeys the exact algebraic identity:

$$4P_{K+2} = 3^{2j+3} + 13 = 243\big(3^{2j-2} - 1\big) + 256.$$

Because $\gcd(243, 256) = 1$, the post-`2` sub-ceiling recovery condition $s_{K+2} \geq 6$ ($64 \mid P_{K+2} \iff 256 \mid 4P_{K+2}$) holds **if and only if**

$$256 \,\Big|\, \big(3^{2j-2} - 1\big) \quad \Longleftrightarrow \quad 64 \,\Big|\, (2j - 2) \quad \big(s_K \equiv 3 \pmod{64}\big).$$

3. **Collapse of the Base Case $j = 1$ ($s_K = 3, b_{K+1} = 9$):** At $j = 1, 3^{2j-2} - 1 = 0$, so $4P_{K+2} = 256 \implies P_{K+2} = 64 = 2^6 \cdot 1$, which has $s_{K+2} = 6$ (even) and $u_{K+2} = 1$—immediately triggering the Even-Rung Collapse at $b_{K+3} = 3^5 = 243$ ($P_{K+3} = 730 \equiv 2 \pmod 8 \implies s_{K+3} = 1$).

3.3 Theorem SB.3: Cross-Window Hankel Determinant Annihilation (`equal_step_window_hankel_determinant_identity_and_bound`)

Across any two block windows $[r_1, r_2]$ and $[r_3, r_4]$ of equal step sum $d = T_{r_2} - T_{r_1} = T_{r_4} - T_{r_3}$, the 2×2 Hankel determinant $\det \begin{pmatrix} P_{r_1} & P_{r_2} \\ P_{r_3} & P_{r_4} \end{pmatrix} = P_{r_1}P_{r_4} - P_{r_2}P_{r_3}$ annihilates the leading $\mathcal{H}_\infty^2 (3/2)^{T_{r_1} + T_{r_4}}$ growth term and satisfies the exact identity

$$2^d\big(P_{r_1}P_{r_4} - P_{r_2}P_{r_3}\big) = P_{r_1}A_{r_3,r_4} - P_{r_3}A_{r_1,r_2},$$

forcing $2^d \mid (P_{r_1}A_{r_3,r_4} - P_{r_3}A_{r_1,r_2})$ and, whenever both windows survive ($2A_{r_1,r_2} < 3^d$ and $2A_{r_3,r_4} < 3^d$), trapping the determinant in the open archimedean interval:

$$-3^d P_{r_3} < 2^{d+1}\big(P_{r_1}P_{r_4} - P_{r_2}P_{r_3}\big) < 3^d P_{r_1}.$$

3.4 Theorem SB.4: Universal Two-Seed First-Divergence Valuation Law (`two_seed_first_divergence_valuation_and_bound`)

Whereas Campaign 2 compared two seeds $b_1 < b_2$ only when they shared the *same* block sequence s, Theorem SB.4 analyzes *arbitrary* seed pairs $b_1 < b_2 \in \mathbb{N}$ at their **first binary parity divergence step** n_0 ($\forall k < n_0$, $\mathrm{orbit}(b_1, k) \bmod 2 = \mathrm{orbit}(b_2, k) \bmod 2$ and $\mathrm{orbit}(b_1, n_0) \bmod 2 \neq \mathrm{orbit}(b_2, n_0) \bmod 2$): - The orbit difference $D_{n_0} = \mathrm{orbit}(b_2, n_0) - \mathrm{orbit}(b_1, n_0)$ is an **odd** positive integer (`D % 2 = 1`), - $2^{n_0} D_{n_0} = 3^{n_0}(b_2 - b_1)$, so $2^{n_0} \mid (b_2 - b_1)$ **and** $2^{n_0+1} \nmid (b_2 - b_1)$: the first divergence time n_0 is **literally the 2-adic valuation** $v_2(b_2 - b_1)$, - Consequently, any two seeds $b_1 < b_2$ with $b_2 - b_1 < 2^N$ **must diverge at some step** $n_0 < N$, and at step n_0 their companion fractional parts split into the disjoint intervals $[0, 1/3)$ (even branch) and $[1/3, 1/2)$ (odd branch).

3.5 Theorem SB.5: Post-Bit-Length Unique Seed Pinning (`post_bitlength_unique_seed_in_dyadic_box`)

Whenever $b_1, b_2 < 2^{T_N + 1}$ both realize an N-block prefix s ($N \geq 1$), $b_1 = b_2$. Thus, once the elapsed binary steps T_N exceed the bit-length W of a fixed seed $b_0 < 2^W$, b_0 is the unique integer in $[0, 2^{T_N+1})$ compatible with s, forcing all top $(T_N + 1 - W)$ binary bits of the 2-adic Hensel lift $(-A_{0,N}(s) \cdot 3^{-T_N}) \bmod 2^{T_N+1}$ to vanish simultaneously.

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4. The Three-Wildcard Expedition: Density-Zero Extinction, Base-3 Frozen Prefixes, and the Mahler–Antihydra $BB(6)$ Duality

In `vibe_mathing_wildcards.py` and Sections 6–8 of `SemigroupBreakoutTheorems.lean`, we extended the breakout campaign along three wildcard directions:

4.1 Theorem SB.6 (Wildcard 1): The 4-Lift Double-Odd Exclusion & $(3/4)^m$ Exponential Density-Zero Mechanism (`dyadic_four_lift_double_odd_bijection_and_three_of_four_survival`)

For any prefix seed $r \in [0, 2^k)$ and 2-bit dyadic lift $b_0 = r + 2^k t$ ($t \in \{0,1,2,3\}$): 1. Whenever the parities match for $j < k$, $\text{orbit}(r + 2^k t, k) = \text{orbit}(r, k) + 3^k t$. 2. A double-odd step occurs at $(k, k+1)$ if and only if $\text{orbit}(b_0, k) \equiv 3 \pmod 4$:

$$\left(\text{orbit}(b_0,k) \bmod 2 = 1 \ \wedge\ \text{orbit}(b_0,k+1) \bmod 2 = 1\right) \iff \text{orbit}(b_0,k) \bmod 4 = 3.$$

3. Because $3^k \bmod 4 \in \{1,3\}$ is a unit modulo 4, as t ranges over $\{0,1,2,3\}$, **exactly 1 of the 4 lifts hits the forbidden $\equiv 3 \pmod 4$ double-odd class and dies**, leaving **at most 3 out of 4 lifts** surviving each 2-step block. Across m disjoint 2-step blocks in $[0, 4^m)$, at most 3^m seeds avoid double-odd steps at even positions (and at most F_{2m+2} avoid all double-odd steps):

m	Horizon $2m$	Box Size 4^m	No-Even-$(1,1)$ Count 3^m	Fibonacci Bound F_{2m+2}	Actual Mahler Survivors	Surviving Fraction
1	2	4	3	3	3	0.750000
2	4	16	9	8	8	0.500000
3	6	64	27	21	18	0.281250
4	8	256	81	55	40	0.156250
5	10	1024	243	144	90	0.087891
6	12	4096	729	377	201	0.049072
7	14	16384	2187	987	452	0.027588

4.2 Theorem SB.7 (Wildcard 2): Raw-Orbit Base-3 "Frozen Top Digits" & Forbidden Trit 2 (`raw_orbit_base3_frozen_prefix_and_leading_trit_le_one`)

Flipping from base 2 to base 3 along the raw binary orbit $2^N g_N = 3^N g_0 + K_N$ ($2K_N < 3^N, 1 \le N$): 1. $\lfloor 2^N g_N / 3^N \rfloor = g_0$ and $(2^N g_N) \bmod 3^N = K_N$: **in base 3, the top digits of $2^N g_N$ above position N are permanently frozen at the initial integer g_0 for all $N \ge 1$.** 2. The $(N-1)$-st base-3 digit of $2^N g_N$ satisfies $\lfloor ((2^N g_N) \bmod 3^N) / 3^{N-1} \rfloor \le 1$ (**never 2**). 3. **$\mathbb{Z}_2$-Symmetry-Breaking Base-3 Length Bound:** When $g_0 < 3^L$ is a true whole number of at most L base-3 digits, $2^N g_N < 3^{N+L}$ has at most $N + L$ base-3 digits.

4.3 Theorem SB.8 (Wildcard 3): Exact Mahler–Antihydra $BB(6)$ Complement Duality (`mahler_antihydra_bb6_complement_duality`)

The famous 6-state Busy Beaver holdout **Antihydra** (`bbchallenge`, 2024) iterates the floor map $a_{k+1} = \lfloor 3a_k/2 \rfloor$ starting at $a_0 = 8$, whereas Mahler's $3/2$ problem iterates the ceiling map $g_{k+1} = \lceil 3g_k/2 \rceil$. We discovered and formalized in Lean 4 an **exact complement duality** linking the two systems at every finite horizon N: - For every horizon $N \in \mathbb{N}$, every Antihydra seed $a_0 \le 2^N$, and every step $0 \le k \le N$, setting the Mahler seed to be the N-bit complement $b_0 := 2^N - a_0$:

$$\boxed{\text{orbit}\left(2^N - a_0,\ k\right) + \text{antihydraOrbit}(a_0, k) = 3^k \cdot 2^{N-k},}$$

and for every $0 \le k < N$:

$$\boxed{\text{orbit}\left(2^N - a_0,\ k\right) \bmod 2 = \text{antihydraOrbit}(a_0, k) \bmod 2}$$

- Thus **Antihydra's binary parity sequence from $a_0 = 8$ is 100% identical to Mahler's binary parity sequence from $b_0 = 2^N - 8$ for the first N steps**, and at steps $k = 11, 12$ (`681 → 1021`), Antihydra hits two consecutive odd steps (`antihydraOrbit 8 11 % 2 = 1 ∧ antihydraOrbit 8 12 % 2 = 1`), meaning its dual Mahler seed $2^N - 8$ dies as a Mahler companion at step 12.

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5. Second-Wave Breakout Theorems: Zero-Carry Base-3 Digit Sums, Late-Step Cauchy Pinch, and Affine $\times(3/2)$ Counter Conjugacy

In `vibe_mathing_ideas_1_2_3.py` and Sections 9–11 of `SemigroupBreakoutTheorems.lean`, we formalized three additional breakout mechanisms:

5.1 Theorem SB.9 (Idea 1): Zero-Carry Base-3 Digit-Sum Conservation & Even-Step Invariance (`zero_carry_base3_digit_sum_and_even_step_invariance`)

Whereas a 2-adic integer $g_0 \in \mathbb{Z}_2$ does not have a finite base-3 digit sum, any whole number $n \in \mathbb{N}$ has a well-defined recursive ternary digit sum $s_3(n) = \text{base3DigitSum}(n)$. Because $3^N g_0$ occupies ternary positions $\ge N$ while $K_N < 3^N$ occupies ternary positions $< N$, adding $3^N g_0 + K_N$ produces **zero base-3 carries**:

$$\boxed{\text{base3DigitSum}\left(2^N g_N\right) = \text{base3DigitSum}(g_0) + \text{base3DigitSum}(K_N) \qquad (\forall N \geq 1).}$$

Furthermore, because $\text{base3DigitSum}(3m) = \text{base3DigitSum}(m)$, on **every even step** ($\text{orbit}(g_0, N) \bmod 2 = 0$, where $2^{N+1}\,\text{orbit}(g_0, N+1) = 3(2^N\,\text{orbit}(g_0, N))$), the base-3 digit sum of $2^N g_N$ is **completely frozen**:

$$\text{orbit}(g_0, N) \bmod 2 = 0 \implies \text{base3DigitSum}\left(2^{N+1}\,\text{orbit}(g_0, N+1)\right) = \text{base3DigitSum}\left(2^N\,\text{orbit}(g_0, N)\right).$$

Since at least $2/3$ of all steps along a surviving Mahler orbit are even, $s_3(2^N g_N)$ is frozen across at least $2/3$ of all steps.

5.2 Theorem SB.10 (Idea 2): Late-Step Cauchy Pinch & Odd-Quotient GCD Divisibility (`late_step_cauchy_pinch_and_odd_quotient_gcd_dvd`)

Eliminating the initial seed P_0 between two late block boundaries N and $N+M$ (with shifted block sequence $s'(t) = s(N+t)$ and step sum $d = T_{N+M} - T_N$) yields: 1. **P_0-Free Late-Step Identity:**

$$2^d P_{N+M} = 3^d P_N + A_{N,N+M} \quad \iff \quad 2^{d+s_{N+M}} u_{N+M} = 3^d \cdot 2^{s_N} u_N + A_{N,N+M}.$$

2. **Common-Factor Divisibility:**

$$\gcd\left(u_N,\, u_{N+M}\right) \;\Big|\; A_{N,N+M}.$$

3. **Exponentially Shrinking Cauchy–Baker Pinch:** Whenever the M-block segment $[N, N+M]$ survives ($0 < \Sigma_M(s')$),

$$0 < 2\left(2^{d+s_{N+M}} u_{N+M} - 3^d \cdot 2^{s_N} u_N\right) = 2A_{N,N+M} < 3^d,$$

so dividing by $3^d P_N$ gives $0 < \frac{P_{N+M}}{P_N}(2/3)^d - 1 < \frac{1}{2P_N} \asymp (2/3)^{T_N} \to 0$.

5.3 Theorem SB.11 (Idea 3): Universal Affine $\times(3/2)$ Counter Conjugacy to Mahler & Closed $BB(6)$ Counter Non-Halting (`affine_counter_mahler_conjugacy_and_closed_bb6_non_halting`)

For any parameter $m \in \mathbb{N}$, the general affine $\times(3/2)$ counter machine

$$C_m(c) := \begin{cases} \lfloor 3c/2 \rfloor + m & \text{if } c \equiv 0 \pmod 2, \\ \lfloor (3c + 2m + 1)/2 \rfloor & \text{if } c \equiv 1 \pmod 2 \end{cases}$$

is **100% conjugate to Mahler's ceiling map** via a constant shift $+2m$:

$$\boxed{\text{affineCounterOrbit}(m, c_0, k) + 2m = \text{orbit}(c_0 + 2m, k) \quad \text{and} \quad \text{affineCounterOrbit}(m, c_0, k) \bmod 2 = \text{orbit}(c_0 + 2m, k) \bmod 2 \quad (\forall k \in \mathbb{N}),}$$

and along the even sub-lattice $4a$, the $BB(6)$ sweep counter `bb6EvenSweepOrbit` preserves evenness (`% 2 = 0`, avoiding the odd halting condition) while growing strictly (`4 * a < 6 * a + 2`).

5.4 Theorem SB.12: The Dual Base-2 / Base-3 Odd-Block Digit-Sum Conservation Tower (`dual_base2_base3_block_digitsum_conservation_and_parity`)

By intersecting the Base-2 Hamming-weight tower $s_2(n) = \text{base2DigitSum}(n)$ with the Base-3 digit-sum tower $s_3(n) = \text{base3DigitSum}(n)$ across the odd-block transition $2^{s_{r+1}} u_{r+1} = 3^{s_r} u_r + 1$ (`vibe_mathing_dual_digitsum_tower.py`), we proved the **Dual Base-2 / Base-3 Conservation Laws**: 1. **Base-2 Division Invariance & Step Law:** Across $P_r = 2^{s_r} u_r$, dividing by 2^{s_r} preserves the binary Hamming weight identically:

$$s_2(P_r) = s_2(u_r), \qquad \text{and} \qquad s_2(u_{r+1}) = s_2\left(3^{s_r} u_r + 1\right).$$

2. **Base-3 Multiplication Invariance & $+1$ Step Law:** Across $b_{r+1} = 3^{s_r - 1} u_r$ and $P_{r+1} = 3b_{r+1} + 1 = 2^{s_{r+1}} u_{r+1}$, multiplying by $3^{s_r - 1}$ preserves the ternary digit sum while $+1$ increments it by exactly 1:

$$\boxed{s_3(b_{r+1}) = s_3(u_r), \qquad s_3(P_r) = s_3(b_r) + 1, \qquad s_3\left(2^{s_{r+1}} u_{r+1}\right) = s_3(u_r) + 1}$$

3. **Base-3 Parity Dichotomy & Non-Vanishing Jump:** Because $s_3(n) \equiv n \pmod 2$, every odd quotient u_r has odd ternary digit sum ($s_3(u_r) \bmod 2 = 1$) while $P_r = 2^{s_r} u_r$ has even ternary digit sum ($s_3(P_r) \bmod 2 = 0$), so $s_3(P_r) \neq s_3(u_r)$ at every block, and the block jumps $\Delta_3(r) := s_3(2^{s_r} u_r) - s_3(u_r)$ telescope across N blocks via:

$$\boxed{s_3(u_N) + \sum_{r=0}^{N} \left(s_3\left(2^{s_r} u_r\right) - s_3(u_r)\right) = s_3(b_0) + N + 1.}$$

5.5 Theorem SB.13: One-Step Base-3 Doubling Identity & Exact 2^k-Weighted Telescoping Law (`one_step_and_weighted_telescoping_base3_carry_identity`)

From our 5-loop unconstrained Python sandbox (`sandbox_symmetry_breaking_loops.py`), applying s_3 to both sides of the raw binary step $2g_{k+1} = 3g_k + \varepsilon_k$ ($\varepsilon_k = g_k \bmod 2 \in \{0, 1\} < 3$) yields the **zero-carry one-step doubling identity**:

$$s_3\left(2g_{k+1}\right) = s_3(g_k) + \varepsilon_k,$$

and therefore the one-step Kummer carry balance:

$$2s_3(g_{k+1}) - s_3(g_k) = \varepsilon_k + \Big(2s_3(g_{k+1}) - s_3(2g_{k+1})\Big), \qquad \text{with } s_3(2g_{k+1}) \le 2s_3(g_{k+1}).$$

Multiplying by 2^k and summing from $k = 0$ to $N-1$ telescopes the left-hand side identically:

$$\boxed{2^N s_3(g_N) = s_3(g_0) + \sum_{k=0}^{N-1} 2^k \varepsilon_k + \sum_{k=0}^{N-1} 2^k \Big(2s_3(g_{k+1}) - s_3(2g_{k+1})\Big) \ge s_3(g_0) + \sum_{k=0}^{N-1} 2^k \varepsilon_k}$$

Notice that $\sum_{k=0}^{N-1} 2^k \varepsilon_k = \text{parityBinaryInt}(g_0, N)$ is **literally the N-bit binary integer whose bits are the parity sequence** $(\varepsilon_0, \ldots, \varepsilon_{N-1})$.

5.6 Theorem SB.14: Top-Down Senge–Straus / Stewart Joint (s_2, s_3) Blueprint (`top_down_senge_straus_stewart_joint_digitsum_blueprint`)

Combining Base-2 division invariance ($s_2(P_r) = s_2(u_r)$) with Base-3 multiplication invariance ($s_3(P_{r+1}) = s_3(u_r) + 1$) proves that the **single odd integer u_r simultaneously controls the Base-2 Hamming weight of P_r and the Base-3 digit sum of P_{r+1}**:

$$\boxed{s_2(P_r) + s_3(P_{r+1}) = s_2(u_r) + s_3(u_r) + 1,}$$

so any Senge–Straus / Stewart joint digit-sum lower bound $M \le s_2(u_r) + s_3(u_r)$ transfers directly to $M + 1 \le s_2(P_r) + s_3(P_{r+1})$, and any lower bound $s_3(g_0) + B < s_3(2^N g_N)$ forces $B < s_3(K_N)$.

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6. Wave-2 Symmetry-Breaking Towers: Legendre Factorial Valuations, the Universal Mod-4 Quotient-Rung Parity Lock, and Weight-2 Domino Collapse

In `sandbox_wave2_towers.py` and Sections 15–17 of `SemigroupBreakoutTheorems.lean`, we built three further $\mathbb{Z}_2$-symmetry-breaking towers:

6.1 Theorem SB.15: Legendre Factorial Valuation $(v_2(n!), v_3(n!))$ Dual Block Tower (`legendre_factorial_valuation_dual_block_tower`)

Using Legendre's exact factorial identities $v_2(n!) + s_2(n) = n$ and $2v_3(n!) + s_3(n) = n$ (which are well-defined only for whole numbers $n \in \mathbb{N}$), we proved that across every odd block r:

$$\boxed{v_2(P_r!) - v_2(u_r!) = \big(2^{s_r} - 1\big)u_r, \qquad 2\Big(v_3(P_{r+1}!) - v_3(u_r!)\Big) = \big(3^{s_r} - 1\big)u_r,}$$

and eliminating u_r yields the **Exact $(2^{s_r} - 1, 3^{s_r} - 1)$ Factorial Cross-Ratio Syzygy**:

$$\boxed{\big(3^{s_r} - 1\big)\Big(v_2(P_r!) - v_2(u_r!)\Big) = 2\big(2^{s_r} - 1\big)\Big(v_3(P_{r+1}!) - v_3(u_r!)\Big).}$$

6.2 Theorem SB.16: Universal Mod-4 Quotient-Rung Parity Lock & Mersenne Trailing-1s Trap (`universal_mod4_quotient_rung_parity_lock_and_mersenne_trap`)

At **every** block r whose successor survives (`2 ≤ s (r + 1)`), the second-lowest binary bit of the odd quotient u_r is 100% locked to the parity of the rung s_r:

$$\boxed{\big(s_r \bmod 2 = 0 \iff u_r \bmod 4 = 3\big) \quad \wedge \quad \big(s_r \bmod 2 = 1 \iff u_r \bmod 4 = 1\big).}$$

Consequently: 1. **Even-Rung Hamming Weight Bound (`2 ≤ s_2(u_r)`):** Whenever s_r is even, $u_r \equiv 3 \pmod 4$ ends in `...11_2`, so $\text{base2DigitSum}(u_r) \ge 2$. 2. **Mersenne Trailing-1s Trap ($u_r \equiv 15 \pmod{16}$):** Whenever $u_r \equiv 15 \pmod{16}$ (ending in four `1`-bits `...1111_2`) at $s_r = 2, P_{r+1} \equiv 8 \pmod{16}$ forces $s_{r+1} = 3 \le 5$, which immediately dies by the post-`2` recovery bound ($s_{r+1} \ge 6$).

6.3 Theorem SB.17: Universal Mod-1024 / Hamming-Weight-2 ($u_K = 1024w + 1$) Three-Block Domino Collapse (`mod1024_and_weight2_odd_quotient_three_block_domino_collapse`)

Generalizing the $u_K = 1$ collapse of Theorem SB.2, we proved that for **every** $u_K = 1024w + 1$ ($u_K \equiv 1 \pmod{1024}$, which includes **every** Hamming-weight-2 odd integer $u_K = 2^m + 1$ with $m \ge 10$): - If $s_K = 2j$ is **even**: $P_{K+1} \equiv 2 \pmod 4$ ($s_{K+1} = 1$, **dies in 1 block**). - If $s_K = 3$: $P_{K+1} = 4(6912w + 7)$ ($s_{K+1} = 2$), $P_{K+2} = 64(972w + 1)$ ($s_{K+2} = 6$ even, $u_{K+2} = 972w + 1 \equiv 1 \pmod 4$), and $P_{K+3} = 708588w + 730 \equiv 2 \pmod 4$ ($s_{K+3} = 1$, **dies in 3 blocks for every** $w \in \mathbb{N}$).

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7. Wave-3 Symmetry-Breaking Towers: The Universal Mod-12 `{5, 7}` Lock, One-Step Backward Quotient Recovery, and the $5 \times / 7 \times$ Sharpened Ceiling

In `sandbox_wave3_towers.py` and Sections 18–20 of `SemigroupBreakoutTheorems.lean`, we combined the Base-2 and Base-3 congruence locks on the odd leftover u_r:

7.1 Theorem SB.18: The Universal Mod-12 `{5, 7}` Lock & Joint Digit-Sum Floor (`universal_mod12_quotient_lock_and_joint_digitsum_ge_five`)

At **every** block $r \geq 0$ whose next block survives (`2 ≤ s (r + 1)`): 1. **Base-3 Ending Lock (from $2^{s_r} u_r = 3b_r + 1 \equiv 1 \pmod 3$):**

$$(s_r \bmod 2 = 0 \iff u_r \bmod 3 = 1) \quad \wedge \quad (s_r \bmod 2 = 1 \iff u_r \bmod 3 = 2).$$

2. **Chinese Remainder Mod-12 `{5, 7}` Lock (combining mod 3 with Theorem SB.16 mod 4):**

$$\boxed{\left(s_r \bmod 2 = 0 \iff u_r \bmod 12 = 7\right) \quad \wedge \quad \left(s_r \bmod 2 = 1 \iff u_r \bmod 12 = 5\right)}$$

Thus, out of the 6 odd residue classes modulo 12 (`1, 3, 5, 7, 9, 11`), **four of them (`1, 3, 9, 11 mod 12`) are 100% extinct at every surviving block**. 3. **Unconditional Exclusion of $u_r = 1$ and $u_r = 3$ & Digit-Sum Floors:** Because $u_r \equiv 5$ or $7 \pmod{12}$ at every surviving block: - $u_r \geq 5$ always ($u_r \geq 7$ whenever s_r is even), and $u_r = 1$ **and** $u_r = 3$ **are impossible for both even and odd** s_r. - $s_2(u_r) \geq 2$ and $s_3(u_r) \geq 2$ at every surviving block, forcing the joint consecutive-block sum to satisfy:

$$\boxed{s_2(P_r) + s_3(P_{r+1}) = s_2(u_r) + s_3(u_r) + 1 \geq 5}$$

7.2 Theorem SB.19: One-Step Backward Quotient Recovery & Mod-3^{s_r} Remainder Law (`one_step_backward_quotient_recovery_and_mod_three_pow_one`)

From $2^{s_{r+1}} u_{r+1} = 3^{s_r} u_r + 1$ and $1 < 3^{s_r}$ ($s_r \geq 1$), integer division by 3^{s_r} recovers the previous odd leftover u_r in one step with remainder identically 1:

$$\boxed{\left\lfloor \frac{2^{s_{r+1}} u_{r+1}}{3^{s_r}} \right\rfloor = u_r \qquad \text{and} \qquad \left(2^{s_{r+1}} u_{r+1}\right) \bmod 3^{s_r} = 1.}$$

7.3 Theorem SB.20: The $5\times$ / $7\times$ Sharpened Global Archimedean Rung Ceiling (`sharpened_five_seven_archimedean_rung_ceiling`)

Injecting $u_K \geq 5$ (and $u_K \geq 7$ when s_K is even) into the Global Archimedean Rung Ceiling (Theorem SB.1) sharpens the ceiling by a factor of $5\times$ (always) and $7\times$ (for even s_K):

$$\boxed{10 \cdot 2^{T_{K+1}} < 3^{T_K}\left(2P_0 + 1\right), \qquad \text{and when } s_K \text{ is even:} \quad 14 \cdot 2^{T_{K+1}} < 3^{T_K}\left(2P_0 + 1\right),}$$

and at $K = 0$, $5 \cdot 2^{s_0} \leq 3b_0 + 1$ ($7 \cdot 2^{s_0} \leq 3b_0 + 1$ when s_0 is even, so $s_0 = 2 \implies b_0 \geq 9$).

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8. Wave-4 Symmetry-Breaking Towers: Two-Step Disjoint Base-3 Digit Blocks, All-K-Step Zero-Carry Top-Block Recovery, and the Two-Sided Past–Future Chinese Remainder Pinch

In `sandbox_wave4_canyon_view.py` and Sections 21–23 of `SemigroupBreakoutTheorems.lean`, we extended the backward quotient recovery tower across 2 steps and across arbitrary M-block windows, and combined the past and future congruences into a two-sided Chinese Remainder pinch:

8.1 Theorem SB.21: Two-Step Disjoint Base-3 Digit-Block Law & Exact Quotient Recovery (`two_step_disjoint_base3_digit_block_law_and_quotient_recovery`)

Across any 2-step block transition $r \to r+1 \to r+2$ (`ValidBlockChain b_0 s u (r + 3)`):

$$2^{s_{r+1}+s_{r+2}} u_{r+2} = 3^{s_r+s_{r+1}} u_r + 3^{s_{r+1}} + 2^{s_{r+1}}.$$

Because $2^{s_{r+1}} < 3^{s_{r+1}}$ and $3^{s_{r+1}} + 2^{s_{r+1}} < 3^{s_r+s_{r+1}}$, the three terms on the right-hand side occupy **completely disjoint Base-3 digit ranges**: - Trits $[0, s_{r+1} - 1]$: the Base-3 digits of $2^{s_{r+1}}$, - Trit s_{r+1}: a single trit `1` from $3^{s_{r+1}}$, - Trits $[s_{r+1} + 1, s_r + s_{r+1} - 1]$: all `0`s, and - Trits $[s_r + s_{r+1}, \infty)$: the Base-3 digits of u_r.

Consequently: 1. **Exact Two-Step Quotient & Remainder Recovery:**

$$\boxed{\left\lfloor \frac{2^{s_{r+1}+s_{r+2}} u_{r+2}}{3^{s_r+s_{r+1}}} \right\rfloor = u_r \qquad \text{and} \qquad \left(2^{s_{r+1}+s_{r+2}} u_{r+2}\right) \bmod 3^{s_r+s_{r+1}} = 3^{s_{r+1}} + 2^{s_{r+1}}.}$$

2. **Universal Zero-Carry Three-Block Base-3 Digit-Sum Splitting:**

$$\boxed{s_3\left(2^{s_{r+1}+s_{r+2}} u_{r+2}\right) = s_3(u_r) + 1 + s_3\left(2^{s_{r+1}}\right).}$$

8.2 Theorem SB.22: All-K-Step Zero-Carry Top-Block Law & Multi-Step Quotient Recovery (`all_k_step_zero_carry_top_block_and_quotient_recovery`)

Across **every** number of future blocks $M \geq 0$ (`ValidBlockChain b_0 s u (r + 1 + M + 1)`) with positive renewal slack $0 < \Sigma_M(s')$ (where $s'(t) = s_{r+1+t}$, $d = \sum_{t=0}^{M-1} s_{r+1+t}$, $S = d + s_{r+1+M}$, $T = s_r + d$, and $R = 3^d + A_{r+1,r+1+M}$): 1. **All-K-Step Identity & Strict Remainder Bound:**

$$2^S u_{r+1+M} = 3^T u_r + R \qquad \text{with} \qquad R < 2 \cdot 3^d < 3^T.$$

2. **Permanent Top-Digit Freezing of u_r Across All Future Blocks:**

$$\boxed{\left\lfloor \frac{2^S u_{r+1+M}}{3^T} \right\rfloor = u_r \qquad \text{and} \qquad \left(2^S u_{r+1+M}\right) \bmod 3^T = R.}$$

3. **All-K-Step Zero-Carry Base-3 Digit-Sum Splitting:**

$$\boxed{s_3\left(2^S u_{r+1+M}\right) = s_3(u_r) + s_3(R).}$$

8.3 Theorem SB.23: Two-Sided Past–Future Chinese Remainder Pinch on Interior Odd Leftovers (`two_sided_past_future_crt_pinch`)

At every interior odd leftover u_{r+1} in a 3-block window (`ValidBlockChain b_0 s u (r + 3)`), u_{r+1} is simultaneously pinched from the **past** (modulo 3^{s_r}) and from the **future** (modulo $2^{s_{r+2}+1}$):

$$\boxed{\left(2^{s_{r+1}} u_{r+1}\right) \bmod 3^{s_r} = 1, \qquad \left(3^{s_{r+1}} u_{r+1} + 1\right) \bmod 2^{s_{r+2}+1} = 2^{s_{r+2}}, \qquad \gcd\left(3^{s_r}, 2^{s_{r+2}+1}\right) = 1,}$$

locking u_{r+1} into a unique Chinese Remainder class modulo $M_{r+1} = 3^{s_r} \cdot 2^{s_{r+2}+1} \geq 72$ whenever $s_r, s_{r+2} \geq 2$.

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9. Wave-5 Symmetry-Breaking Towers: The Unconditional $4/5$ Renewal Ceiling (Slack-Free), $9/5$ All-K-Step Recovery, and the Sliding Cut-Point Past–Future Chinese Remainder Pinch

In `sandbox_wave5_multi_block_crt.py` and Sections 24–26 of `SemigroupBreakoutTheorems.lean`, our Python sandbox revealed that even when $2A_{0,r} > 3^{T_r}$ (e.g., $s = (2, 2)$ where $A_{0,2}/3^4 = 52/81 \approx 0.642 > 1/2$), the geometric sum $\sum_{j=0}^{r-1} (2/3)^{T_{j+1}}$ for surviving blocks ($s_k \geq 2$) is bounded by $\sum_{j=1}^{r} (4/9)^j < 4/5 < 1$. Formalizing this eliminated the sub-ceiling slack hypothesis (`0 < renewalIntSlack`) across the entire multi-block tower:

9.1 Theorem SB.24: The Unconditional $4/5$ Renewal Ceiling & Past Base-3 Recovery (`unconditional_four_fifths_renewal_ceiling_and_past_base3_recovery`)

For **every** surviving block chain (`ValidBlockChain b_0 s u (r + 1)` with $\forall k < r,\ 2 \leq s_k$, **with no `0 < renewalIntSlack` hypothesis needed**): 1. **Exact $4/5$ Inductive Invariant:**

$$\boxed{5A_{0,r} + 4 \cdot 2^{T_r} \leq 4 \cdot 3^{T_r} \qquad \Longrightarrow \qquad 5A_{0,r} < 4 \cdot 3^{T_r} \quad \wedge \quad A_{0,r} < 3^{T_r}.}$$

2. **Unconditional Past Base-3 Quotient, Remainder & Zero-Carry Digit-Sum Law:**

$$\boxed{\left\lfloor \frac{2^{T_r} P_r}{3^{T_r}} \right\rfloor = P_0, \qquad \left(2^{T_r} P_r\right) \bmod 3^{T_r} = A_{0,r}, \qquad s_3\left(2^{T_r} P_r\right) = s_3(P_0) + s_3(A_{0,r}).}$$

9.2 Theorem SB.25: Unconditional All-K-Step Odd-Leftover Recovery & $9/5$ Geometric Remainder Bound (`unconditional_all_k_step_odd_leftover_recovery_and_nine_fifths_bound`)

For **every** surviving block chain (`ValidBlockChain b_0 s u (r + 1 + M + 1)` with $\forall k < M,\ 2 \leq s_{r+1+k}$, **eliminating `0 < renewalIntSlack` from Theorem SB.22**), with $s'(t) = s_{r+1+t}$, $d = \sum_{t=0}^{M-1} s_{r+1+t}$, $S = d + s_{r+1+M}$, $T = s_r + d$, and $R = 3^d + A_{r+1,r+1+M}$: 1. **Universal $9/5$ Geometric Remainder Bound:**

$$\boxed{5R + 4 \cdot 2^d \leq 9 \cdot 3^d \qquad \Longrightarrow \qquad R < 2 \cdot 3^d < 3^T.}$$

2. **Unconditional Multi-Step Quotient, Remainder & Zero-Carry Digit-Sum Law:**

$$\boxed{\left\lfloor \frac{2^S u_{r+1+M}}{3^T} \right\rfloor = u_r, \qquad \left(2^S u_{r+1+M}\right) \bmod 3^T = R, \qquad s_3\left(2^S u_{r+1+M}\right) = s_3(u_r) + s_3(R).}$$

9.3 Theorem SB.26: Sliding Cut-Point Past–Future Chinese Remainder Pinch & Post-Bit-Length Uniqueness (`sliding_cut_point_past_future_crt_pinch`)

Across any $(r + M + 1)$-block chain (`ValidBlockChain b_0 s u (r + M + 1)` with $\forall k < r,\ 2 \leq s_k$), at any interior cut point r with past step sum $T_r = \sum_{k=0}^{r-1} s_k$, future interior step sum $d_r = \sum_{t=0}^{M-1} s_{r+t}$, total future halvings $D_r = d_r + s_{r+M}$, and combined Past-Base-3 $\times$ Future-Base-2 modulus $\mathcal{M}_r = 3^{T_r} \cdot 2^{D_r+1}$: 1. **Simultaneous Coprime Past Base-3 & Future Base-2 Congruences at Cut Point r:**

$$\boxed{\left(2^{T_r} P_r\right) \bmod 3^{T_r} = A_{0,r}, \qquad \left(3^{d_r} P_r + A_{r,r+M}\right) \bmod 2^{D_r+1} = 2^{D_r}, \qquad \gcd\left(3^{T_r}, 2^{D_r+1}\right) = 1.}$$

2. **Unconditional Sliding Cut-Point Archimedean Bound & Post-Bit-Length Pinch:**

$$2^{T_r} P_r < 3^{T_r}(P_0 + 1) \qquad \text{and} \qquad 5 \cdot 2^{T_r + D_r} P_r < 3^{T_r} \cdot 2^{D_r}(5P_0 + 4),$$

so whenever $P_0 < 2^{T_r + D_r + 1}$ (the post-bit-length horizon), at **every** cut point r:

$$\boxed{P_r < 3^{T_r} \cdot 2^{D_r+1} = \mathcal{M}_r}$$

9.4 Theorem SB.27: The Zero-Carry Base-3 Horner Remainder Ladder (`zero_carry_base3_horner_remainder_ladder`)

Peeling the cumulative remainder $R = 3^d + A_{r+1,r+1+M}$ one level deeper using Theorem SB.24 ($A_{r+1,r+1+M} < \frac{4}{5} \cdot 3^d < 3^d$) proves that R itself splits in Base 3 with **zero carries**:

$$\boxed{\left\lfloor \frac{R}{3^d} \right\rfloor = 1, \qquad R \bmod 3^d = A_{r+1,r+1+M}, \qquad s_3(R) = 1 + s_3\big(A_{r+1,r+1+M}\big),}$$

giving a **three-zone zero-carry Base-3 decomposition** across every multi-step window M:

$$\boxed{s_3\big(2^S u_{r+1+M}\big) = s_3(u_r) + 1 + s_3\big(A_{r+1,r+1+M}\big)}$$

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10. Wave-6 Symmetry-Breaking Towers: Exploring Bridge A ($\mathbb{R} \times \mathbb{Q}_2$ Dual-Universe Series) and Bridge B (Additive Carry-Isolation & Horner Transducer)

In `sandbox_wave6_both_bridges.py` and Sections 28–30 of `SemigroupBreakoutTheorems.lean`, we formalized both candidate bridges across the middle of the canyon:

10.1 Theorem SB.28 (Bridge A): Dual-Universe Series Lowest-Terms & $\mathbb{R} \times \mathbb{Q}_2$ Uniqueness of P_0 (`bridge_a_dual_universe_series_lowest_terms_and_uniqueness`)

At every surviving block $K \geq 0$ (`ValidBlockChain b_0 s u (K + 1)` with $\forall k < K,\ 2 \leq s_k$), consider the rational approximant

$$Q_K := P_0 + \sum_{j=1}^{K} \left(\frac{2}{3}\right)^{T_j} = \frac{3^{T_K} P_0 + A_{0,K}}{3^{T_K}} = \frac{2^{T_{K+1}} u_K}{3^{T_K}}$$

of the dual-universe lacunary series $F(2/3) = \sum_{j=1}^{\infty} (2/3)^{T_j}$: 1. **Strict Lowest Terms & Exact Weil Height:**

$$\boxed{\gcd\left(2^{T_{K+1}} u_K,\ 3\right) = 1 \qquad \text{and} \qquad \gcd\left(2^{T_{K+1}} u_K,\ 3^{T_K}\right) = 1,}$$

so $Q_K = (2^{T_{K+1}} u_K)/3^{T_K}$ is in **strict lowest terms** at every block K. 2. **Simultaneous Archimedean ($\mathbb{R}$) and 2-Adic ($\mathbb{Q}_2$) Pinning of P_0: - In $\mathbb{R}$:** $\lfloor 2^{T_{K+1}} u_K / 3^{T_K} \rfloor = P_0$ and $(2^{T_{K+1}} u_K) \bmod 3^{T_K} = A_{0,K}$. - **In $\mathbb{Q}_2$:** $(3^{T_K} P_0 + A_{0,K}) \bmod 2^{T_{K+1}+1} = 2^{T_{K+1}}$ (exact 2-adic distance $|F_K(2/3) - (-P_0)|_2 = 2^{-T_{K+1}}$), and in the post-bit-length regime ($P_0 < 2^{T_{K+1}}$), for **any** candidate $P_0' < 2^{T_{K+1}}$ satisfying $(3^{T_K} P_0' + A_{0,K}) \bmod 2^{T_{K+1}} = 0$, we have $P_0' = P_0$ **identically**.

10.2 Theorem SB.29 (Bridge B1): Seed-Independent Step-to-Step Base-3 Evolution Law (`bridge_b_seed_independent_base3_step_evolution`)

Across any `ValidBlockChain b_0 s u (K + 1)` with $\forall k < K,\ 2 \leq s_k$: 1. **Unconditional Odd-Leftover Base-3 Law:**

$$\boxed{s_3\big(2^{T_{r+1}} u_r\big) = s_3(P_0) + s_3(A_{0,r}) \qquad \text{for all } r \leq K.}$$

2. **Seed-Independent Two-Block Base-3 Evolution:** For **any** two blocks $r_1, r_2 \leq K$:

$$\boxed{s_3\big(2^{T_{r_2+1}} u_{r_2}\big) + s_3(A_{0,r_1}) = s_3\big(2^{T_{r_1+1}} u_{r_1}\big) + s_3(A_{0,r_2}),}$$

proving that the change in $s_3(2^{T_{r+1}} u_r)$ between any two blocks r_1 and r_2 is **100% independent of the initial seed b_0 (and P_0)** and depends **only** on the step sequence s.

10.3 Theorem SB.30 (Bridge B2): Additive Carry-Isolation Law Across the Multi-Step Boundary (`bridge_b_additive_carry_isolation_and_horner_step`)

For $X = 2^S u_{r+1+M} = 3^T u_r + R$ (where $T = s_r + d$ and $R = 3^d + A_{r+1,r+1+M}$): Because $5R < 9 \cdot 3^d$ (`SB.25`) and $9 \cdot 2^{s_r} \leq 4 \cdot 3^{s_r}$ ($s_r \geq 2$), even **after** multiplying X by 2^{s_r}, the scaled remainder $2^{s_r} R$ satisfies:

$$\boxed{5 \cdot \left(2^{s_r} R\right) < 4 \cdot 3^T \qquad \Longrightarrow \qquad 2^{s_r} R < 3^T}$$

Therefore, $2^{s_r} R$ **still never carries** into $3^T (2^{s_r} u_r)$ in Base 3:

$$\boxed{s_3\big(2^{s_r} X\big) = s_3\big(2^{s_r} u_r\big) + s_3\big(2^{s_r} R\big),}$$

and the Base-3 carry defect $\Delta_{2^{s_r}}(n) := s_3(2^{s_r} n) - s_3(n)$ splits **additively with zero cross-talk** across the multi-step boundary:

$$\boxed{s_3\big(2^{s_r} X\big) + s_3(u_r) + s_3(R) = s_3(X) + s_3\big(2^{s_r} u_r\big) + s_3\big(2^{s_r} R\big)}$$

11. Wave-7 Symmetry-Breaking Towers (Bridge C): The Binary Trailing-Ones Depletion & Halving-Budget Bridge

In `sandbox_wave7_binary_carry_bridge.py` and Sections 31–33 of `SemigroupBreakoutTheorems.lean`, we discovered the **Base-2 counterpart** to the Base-3 Zero-Carry Firewall (`SB.24–SB.30`):

11.1 Theorem SB.31 (Bridge C1): One-Step Binary Trailing-Ones Depletion Law (`bridge_c_one_step_binary_trailing_ones_depletion_law`)

At every block r, because $3^{s_r}u_r + 1 = 2^{s_{r+1}}u_{r+1}$ and u_{r+1} is odd:

$$3^{s_r}u_r = 2^{s_{r+1}}\big(u_{r+1} - 1\big) + \big(2^{s_{r+1}} - 1\big).$$

In Base 2, the bottom s_{r+1} bits of $3^{s_r}u_r$ are **all `1`s** ($2^{s_{r+1}} - 1$, with binary digit sum s_{r+1}), and the bits above $2^{s_{r+1}}$ are $u_{r+1} - 1$ (with binary digit sum $s_2(u_{r+1}) - 1$ since u_{r+1} is odd). Consequently: 1. **Exact Base-2 Quotient & All-Ones Remainder:**

$$\boxed{\left\lfloor \frac{3^{s_r}u_r}{2^{s_{r+1}}} \right\rfloor = u_{r+1} - 1 \qquad \text{and} \qquad \big(3^{s_r}u_r\big) \bmod 2^{s_{r+1}} = 2^{s_{r+1}} - 1.}$$

2. **Exact Binary Carry-Depletion Law:**

$$\boxed{s_2\big(3^{s_r}u_r\big) + 1 = s_2(u_{r+1}) + s_{r+1}}$$

In words: adding $+1$ to $3^{s_r}u_r$ consumes **exactly s_{r+1} trailing binary `1`-bits** and replaces them with a single `1`-bit at position s_{r+1}.

11.2 Theorem SB.32 (Bridge C2): Multi-Step Dual Base-2 / Base-3 Zero-Carry Split of N_K (`bridge_c_multi_step_dual_base2_base3_zero_carry_split`)

For the multi-block quantity $N_K := 3^{T_K}P_0 + A_{0,K} = 2^{T_{K+1}}u_K$: 1. **In Base 3 ($N_K$ splits at 3^{T_K} with zero carries):**

$$\boxed{\left\lfloor \frac{N_K}{3^{T_K}} \right\rfloor = P_0, \qquad N_K \bmod 3^{T_K} = A_{0,K}, \qquad s_3(N_K) = s_3(P_0) + s_3(A_{0,K}).}$$

2. **In Base 2 ($N_K - 1$ splits at $2^{T_{K+1}}$ with zero carries):**

$$\boxed{\left\lfloor \frac{N_K - 1}{2^{T_{K+1}}} \right\rfloor = u_K - 1, \qquad (N_K - 1) \bmod 2^{T_{K+1}} = 2^{T_{K+1}} - 1, \qquad s_2(N_K - 1) + 1 = T_{K+1} + s_2(u_K)}$$

11.3 Theorem SB.33 (Bridge C3): The Telescoped Binary Halving-Budget Law (`bridge_c_telescoped_binary_halving_budget_law`)

Telescoping Theorem SB.31 across K blocks ($r = 0, \ldots, K-1$) yields the **Exact K-Block Binary Halving-Budget Law**:

$$\boxed{\big(T_{K+1} - s_0\big) + s_2(u_K) + \sum_{r=0}^{K-1} s_2(u_r) = s_2(u_0) + K + \sum_{r=0}^{K-1} s_2\big(3^{s_r}u_r\big)}$$

In plain English: every single halving step in $T_{K+1} - s_0 = s_1 + \cdots + s_K$ (beyond $+1$ per block) **must be paid for** out of the initial binary `1`-bits $s_2(u_0)$ and the net binary `1`-bits manufactured by multiplying u_r by 3^{s_r}.

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12. Overnight 20-Loop Bridge Exploration Campaign (Loops 1–20: Theorems `SB.34–SB.53`)

In `sandbox_overnight_20_bridge_loops.py` and Sections 34–53 of `SemigroupBreakoutTheorems.lean`, we executed a 20-loop campaign across four 5-loop arcs:

12.1 Arc I (Loops 1–5, Theorems `SB.34–SB.38`): Kummer Carry & Multi-Block Factorial Valuation Bridges

1. **Theorem SB.34 (`kummer_legendre_one_step_binary_factorial_bridge`):**

$$v_2\big((3^{s_r}u_r)!\big) + s_{r+1} = \big(2^{s_{r+1}} - 1\big)u_{r+1} + v_2(u_{r+1}!) = v_2(P_{r+1}!).$$

2. **Theorem SB.35 (`zero_carry_three_adic_factorial_and_binomial_valuation_law`):** For $N_K = 3^{T_K}P_0 + A_{0,K} = 2^{T_{K+1}}u_K$:

$$\boxed{v_3(N_K!) = v_3\big((3^{T_K}P_0)!\big) + v_3(A_{0,K}!) \qquad \Longleftrightarrow \qquad v_3\left(\binom{3^{T_K}P_0 + A_{0,K}}{A_{0,K}}\right) = 0,}$$

   together with $v_2(N_K!) + s_2(u_K) = N_K$ and $v_2((N_K - 1)!) + T_{K+1} + s_2(u_K) = N_K$.

3. **Theorem SB.36 (`joint_multi_block_factorial_valuation_syzygy`):**

$$\boxed{v_2(N_K!) + s_2(u_K) = 2v_3(N_K!) + s_3(P_0) + s_3(A_{0,K}).}$$

4. **Theorem SB.37 (`kummer_base3_two_pow_mul_carry_identity_and_bound`):** For every $s, m \in \mathbb{N}$:

$$\boxed{s_3(2^s m) + 2\big(v_3((2^s m)!) - 2^s v_3(m!)\big) = 2^s s_3(m), \qquad s_3(2^s m) \le 2^s s_3(m).}$$

5. **Theorem SB.38 (`multi_step_base3_horner_digitsum_lower_envelope`):**

$$s_3\big(2^S u_{r+1+M}\big) = s_3(u_r) + 1 + s_3\big(A_{r+1,r+1+M}\big) \ge 3 \quad (\ge 4 \text{ whenever } M \ge 1).$$

12.2 Arc II (Loops 6–10, Theorems `SB.39–SB.43`): Mod-36, Mod-24, Mod-16 & $79\times$ Interior Floor Bridges

1. **Theorem SB.39 (`interior_mod36_rung_to_quotient_bijection`):** At every interior block $r+1$ ($s_r \ge 2, s_{r+2} \ge 2$), $(2^{s_{r+1}} u_{r+1}) \bmod 9 = 1$, locking $u_{r+1} \bmod 36$ into a 1-to-1 bijection with $s_{r+1} \bmod 6$:

$$\boxed{s_{r+1} \bmod 6 \in \{0,1,2,3,4,5\} \iff u_{r+1} \bmod 36 \in \{19,5,7,17,31,29\}}$$

   Thus **12 of the 18 odd residue classes modulo 36 are 100% extinct** at every interior surviving block.

2. **Theorem SB.40 (`interior_extinction_of_five_and_seven_and_seventeen_floor`):** At every interior block $r+1$ surviving two steps ($s_r, s_{r+1}, s_{r+2}, s_{r+3} \ge 2$):
3. $u_{r+1} = 5 \implies s_{r+1} \ge 7$ (extinct for $2 \le s_{r+1} \le 6$),
4. $\neg(s_{r+1} = 2 \wedge u_{r+1} = 7)$ (since $3^2 \cdot 7 + 1 = 64 = 2^6 \cdot 1 \implies u_{r+2} = 1$, dying at block $r+3$),
5. $u_{r+1} = 7 \implies s_{r+1} \ge 8$ (extinct for $2 \le s_{r+1} \le 7$), and
6. $s_{r+1} \le 6 \implies u_{r+1} \ge 17$ ($s_{r+1} = 2 \implies u_{r+1} \ge 43$).
7. **Theorem SB.41 (`even_rung_mod16_trap_and_seventy_nine_floor_at_rung_two`):** At $s_{r+1} = 2$, $u_{r+1} \bmod 16 = 11$ forces $s_{r+2} = 2$ and $u_{r+2} \bmod 4 = 1$, dying at block $r+3$. Since $43 \bmod 16 = 11$, $u_{r+1} = 43$ also dies in 2 blocks, forcing:

$$\boxed{s_{r+1} = 2 \implies u_{r+1} \ge 79}$$

8. **Theorem SB.42 (`universal_mod8_and_mod24_next_rung_dichotomy`):** The third bit of u_r ($u_r \bmod 8$, and $u_r \bmod 24$) universally determines whether the next block has $s_{r+1} = 2$ or $s_{r+1} \ge 3$:
9. Even s_r: $(u_r \bmod 8 = 3 \iff u_r \bmod 24 = 19 \iff s_{r+1} = 2)$ and $(u_r \bmod 8 = 7 \iff u_r \bmod 24 = 7 \iff s_{r+1} \ge 3)$.
10. Odd s_r: $(u_r \bmod 8 = 1 \iff u_r \bmod 24 = 17 \iff s_{r+1} = 2)$ and $(u_r \bmod 8 = 5 \iff u_r \bmod 24 = 5 \iff s_{r+1} \ge 3)$.
11. **Theorem SB.43 (`sharpened_seventeen_seventy_nine_interior_archimedean_ceiling`):** At every interior block $r+1$ ($s_k \ge 2$):

$$\boxed{s_{r+1} \le 6 \implies 85 \cdot 2^{T_{r+2}} < 3^{T_{r+1}}(5P_0 + 4), \qquad s_{r+1} = 2 \implies 395 \cdot 2^{T_{r+2}} < 3^{T_{r+1}}(5P_0 + 4)}$$

12.3 Arc III (Loops 11–15, Theorems `SB.44–SB.48`): Wronskian, Cross-Ratio & Multi-Window Determinant Bridges

1. **Theorem SB.44 (`adjacent_block_wronskian_and_equal_rung_log_concavity`):**

$$2^{s_r} \cdot 3^{s_{r+1}} P_{r+1}(P_{r+1} - 1) = 3^{s_r} \cdot 2^{s_{r+1}} P_r(P_{r+2} - 1),$$

   and whenever $s_r = s_{r+1}$, $P_{r+1}^2 - P_r P_{r+2} = P_{r+1} - P_r > 0$ (strict log-concavity on equal-rung pairs).

2. **Theorem SB.45 (`two_block_odd_leftover_cross_ratio_invariant`):**

$$3^{s_{r_1}} \cdot 2^{s_{r_2}} P_{r_1}(P_{r_2+1} - 1) = 2^{s_{r_1}} \cdot 3^{s_{r_2}} P_{r_2}(P_{r_1+1} - 1).$$

3. **Theorem SB.46 (`three_point_seed_free_linear_eliminator_and_cocycle`):**

$$A_{0,r+M} = 3^d A_{0,r} + 2^{T_r} A_{r,r+M} = 3^d A_{0,r} + 2^{T_r}\big(2^d P_{r+M} - 3^d P_r\big).$$

4. **Theorem SB.47 (`equal_step_odd_leftover_cross_window_determinant_identity`):**

$$2^d\big(2^{s_1+s_4} u_1 u_4 - 2^{s_2+s_3} u_2 u_3\big) = 2^{s_1} u_1 A_{34} - 2^{s_3} u_3 A_{12}.$$

5. **Theorem SB.48 (`unconditional_four_fifths_cross_window_hankel_sandwich`):**

$$-4 \cdot 3^d P_3 < 5 \cdot 2^d (P_1 P_4 - P_2 P_3) < 4 \cdot 3^d P_1.$$

12.4 Arc IV (Loops 16–20, Theorems `SB.49–SB.53`): Base-2/Base-3 Complement Duality & Grand 53-Bridge Synthesis

1. **Theorem SB.49 (`mahler_antihydra_base2_complement_digitsum_law`):** For all $m < 2^N$, $s_2(2^N - 1 - m) + s_2(m) = N$, and for all odd $a_0 < 2^N$, $s_2(2^N - a_0) + s_2(a_0) = N + 1$.
2. **Theorem SB.50 (`mahler_antihydra_base3_complement_digitsum_law`):** For all $m < 3^N$, $s_3(3^N - 1 - m) + s_3(m) = 2N$, and at step $k = N$ of the Mahler–Antihydra duality ($b_N + a_N = 3^N$), $s_3(b_N - 1) + s_3(a_N) = 2N$.
3. **Theorem SB.51 (`post_bitlength_colliding_binary_ternary_window_on_N_K`):** Packages the simultaneous $\mathbb{R} \times \mathbb{Q}_2 \times \mathbb{Q}_3$ pinning and dual Base-2/Base-3 zero-carry splits on N_K with $s_2(N_K - 1) \ge T_{K+1} + 1$.
4. **Theorem SB.52 (`interior_multi_block_joint_digitsum_and_seventy_nine_floor`):** Combines the consecutive-block joint digit-sum floor $s_2(P_{r+1}) + s_3(P_{r+2}) \ge 5$, seed-independent Base-3 evolution, and the $17 \times /79\times$ interior floors.
5. **Theorem SB.53 (`grand_semigroup_breakout_53_bridge_synthesis`):** The Grand Synthesis of all 53 $\mathbb{Z}_2$-symmetry-breaking theorems (`SB.1–SB.53`).

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13. Cross-Application to Collatz ($3n+1$) & Universal Structures in $3/2 \bmod 1$ (Theorems `SB.54–SB.56`)

In Sections 54–56 of `SemigroupBreakoutTheorems.lean`, we cross-applied our Mahler Base-2/Base-3 bridges directly to the Collatz Syracuse map ($2^s u' = 3u+1$) and proved/disproved universal structures in $3/2 \bmod 1$:

13.1 Theorem SB.54 (`collatz_syracuse_bridge_and_one_two_cycle_exclusion`)

The Collatz Syracuse odd-step equation $2^s u' = 3u+1$ is the exact $s_r = 1$ specialization of the Mahler odd-block equation $2^{s_{r+1}} u_{r+1} = 3^{s_r} u_r + 1$. Consequently: 1. **Collatz One-Step Base-3 & Base-2 Carry Laws:** Whenever $2^s u' = 3u+1$ with u' odd:

$$\boxed{\left\lfloor \frac{2^s u'}{3} \right\rfloor = u, \qquad (2^s u') \bmod 3 = 1, \qquad s_3(2^s u') = s_3(u) + 1, \qquad s_2(3u) + 1 = s_2(u') + s}$$

Summing around any closed K-step Collatz loop ($u_K = u_0$ with total halvings $T_K = \sum_{r=0}^{K-1} s_r$) yields the **Collatz Loop Carry-Budget Laws**:

$$\sum_{r=0}^{K-1} \big(s_2(3u_r) - s_2(u_r)\big) = T_K - K \qquad \text{and} \qquad \sum_{r=0}^{K-1} \big(s_3(2^{s_r} u_{r+1}) - s_3(u_{r+1})\big) = K.$$

2. **Unconditional Disproof of Non-Trivial Collatz 1-Cycles and 2-Cycles (Contrasted with $3n-1$):** - For $K=1$: $2^s u_0 = 3u_0 + 1$ ($u_0 \geq 1$) forces $s = 2$ and $u_0 = 1$ ($1 \to 4 \to 2 \to 1$ is the unique 1-cycle). - For $K = 2$: there do **not** exist integers $1 \leq u_0 < u_1$ and exponents $1 \leq s_0, 1 \leq s_1$ satisfying $2^{s_0} u_1 = 3u_0 + 1$ and $2^{s_1} u_0 = 3u_1 + 1$ (because $u_0 < u_1 \implies 2^{s_0} u_1 = 3u_0 + 1 < 4u_1 \implies s_0 = 1 \implies (2^{1+s_1} - 9) u_0 = 5$, which has no integer solution), whereas in $3n-1$, Catalan's $9 - 2^3 = 1$ yields the genuine 2-cycle $5 \to 7 \to 5$.

13.2 Theorem SB.55 (`universal_constant_rung_affine_invariance_and_run_length_bound`)

Across any run of m consecutive blocks with constant halving count $s \geq 1$ ($s_{r+k} = s$ for $0 \leq k \leq m$):

$$\boxed{2^{sm}\big((3^s - 2^s)u_{r+m} + 1\big) = 3^{sm}\big((3^s - 2^s)u_r + 1\big) \implies 2^{sm} \,\Big|\, \big((3^s - 2^s)u_r + 1\big) \implies 2^{sm} \leq (3^s - 2^s)u_r + 1}$$

In particular, for a run of $s = 2, 4^m \mid (5u_r + 1)$ and $4^m \leq 5u_r + 1$, proving universally that **no integer seed can sustain an infinite constant-rung run**.

13.3 Theorem SB.56 (`sharp_four_fifths_equality_characterization_and_two_two_escape`)

1. **Proved Universal Structure:** In the unconditional $4/5$ renewal ceiling $5A_{0,r} + 4 \cdot 2^{T_r} \leq 4 \cdot 3^{T_r}$ (`SB.24`), **exact equality holds for all $k < r$ if $s_k = 2$**, whereas **strict inequality $< 4 \cdot 3^{T_r}$ holds** as soon as a block has $s_k \geq 3$.
2. **Disproved Naive Conjecture:** For $s = (2,2)$, every block satisfies $s_k \geq 2$, yet $2A_{0,2} = 104 > 81 = 3^{T_2}$ (`renewalIntSlack s 2 = -23 < 0`), disproving the naive conjecture that $s_k \geq 2$ automatically implies sub-$1/2$ renewal slack $A_{0,r}/3^{T_r} < 1/2$.

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14. The $(j+n, k+n)$ Diagonal Limit & The $j \geq k$ / $j \gg k$ Regime (Theorems `SB.57–SB.60`)

In Sections 57–60 of `SemigroupBreakoutTheorems.lean` (verified computationally across 122,979 configurations in `sandbox_wave8_jk_diagonal_asymptotics.py`), we formalized the exact behavior of the Mahler orbit along the diagonal exponent shift $(j,k) \longrightarrow (j+n, k+n)$ and in the 3-heavy / 2-starved regime $j \geq k$ and $j \gg k$:

14.1 Theorem SB.57 (`two_window_jk_diagonal_cauchy_tail_freeze_and_quotient_invariance`)

Starting at block r with 3-exponent $j = s_r$, advance across two consecutive future windows of M_1 blocks (n_1 steps, ending rung $k_1 = s_{r+1+M_1}$) and M_2 blocks (n_2 steps, ending rung $k_2 = s_{r+1+M_1+M_2}$), with $S_1 = n_1 + k_1$, $S_{12} = n_1 + n_2 + k_2$, $T_1 = j + n_1$, and $T_{12} = j + n_1 + n_2$: 1. **Exact Two-Window Diagonal Transition:**

$$\boxed{2^{n_1+n_2+k_2} u_{r+1+M_1+M_2} = 3^{n_2}\big(2^{n_1+k_1} u_{r+1+M_1}\big) + 2^{n_1} A^{(2)} = 3^{j+n_1+n_2} u_r + R_{12},}$$

where $R_{12} = 3^{n_2} R_1 + 2^{n_1} A^{(2)}$. 2. **Unconditional $4/5$ Cauchy Tail-Freeze Bound:**

$$\boxed{5\big(R_{12} - 3^{n_2} R_1\big) + 4 \cdot 2^{n_1+n_2} \leq 4 \cdot 2^{n_1} \cdot 3^{n_2} \qquad \text{and} \qquad 5R_{12} + 4 \cdot 2^{n_1+n_2} \leq 9 \cdot 3^{n_1+n_2}.}$$

Dividing by the main term $3^{j+n_1+n_2} u_r$ shows that the relative tail drift between horizon n_1 and horizon n_1+n_2 is bounded by $\frac{4}{5 \cdot 3^j u_r} \left(\frac{2}{3}\right)^{n_1}$, decaying exponentially in **both** j and n_1. 3. **Two-Horizon Quotient & Base-3 Top-Block Invariance:**

$$\left\lfloor \frac{2^{n_1+n_2+k_2} u_{r+1+M_1+M_2}}{3^{j+n_1+n_2}} \right\rfloor = u_r = \left\lfloor \frac{2^{n_1+k_1} u_{r+1+M_1}}{3^{j+n_1}} \right\rfloor \quad \text{and} \quad s_3\big(2^{S_{12}} u_{r+1+M_1+M_2}\big) = s_3(u_r) + s_3(R_{12}).$$

14.2 Theorem SB.58 (`j_ge_k_excess_explosion_binary_swelling_and_halving_debt_law`)

Whenever a block transition $2^k u_{r+1} = 3^j u_r + 1$ has $j = s_r \geq k = s_{r+1} \geq 2$ with excess $\Delta = j - k \geq 0$: 1. **Excess-Exponent Decomposition & Super-Exponential Floor Jump:**

$$\boxed{2^k u_{r+1} = 3^\Delta \cdot 3^k u_r + 1, \qquad 4u_{r+1} \geq 9 \cdot 3^\Delta u_r + 1, \qquad u_{r+1} > 2 \cdot 3^\Delta u_r.}$$

2. **Binary 1-Bit Budget Law Under $j \geq k$:**

$$s_2(u_{r+1}) + k = s_2(3^j u_r) + 1.$$

3. **Two-Step Halving-Debt Crisis (Return-to-Baseline Barrier):** Across block r ($j = s_r, k = s_{r+1}$) and block $r+1$ ($k = s_{r+1}, k' = s_{r+2}$):

$$\boxed{2^{k+k'} u_{r+2} = 3^{j+k} u_r + 3^k + 2^k, \qquad \text{so} \qquad u_{r+2} \leq u_r \implies 2^{k+k'} > 3^{j+k} \ \wedge \ 2^{k'} > 3^j}$$

14.3 Theorem SB.59 (`three_adic_seed_diagonal_shift_and_dead_zone_invariance_law`)

Whenever the initial return integer $P_0 = 3^{j_0} m$ (or any interior odd core $u_r = 3^{j_0} m$) contains a 3-adic factor 3^{j_0}, advancing $n = T_K$ steps with ending rung $k = s_K$ shifts the exponents along the exact diagonal $(j_0, k) \longrightarrow (j_0 + n, k + n)$:

$$\boxed{N_K = 2^{n+k} u_K = 3^{j_0+n} m + A_{0,K}, \qquad \left\lfloor \frac{N_K}{3^{j_0+n}} \right\rfloor = m, \qquad N_K \bmod 3^{j_0+n} = A_{0,K}.}$$

In addition, N_K exhibits an **exact j_0-trit Dead Zone (`00...0` in Base 3)** in $[3^n, 3^{j_0+n})$ and complete **Base-3 digit-sum invariance in j_0**:

$$\boxed{\left\lfloor \frac{N_K}{3^n} \right\rfloor \bmod 3^{j_0} = 0, \qquad \left\lfloor \frac{N_K \bmod 3^{j_0+n}}{3^n} \right\rfloor = 0, \qquad s_3\big(2^{n+k} u_K\big) = s_3(m) + s_3(A_{0,K})}$$

14.4 Theorem SB.60 (`grand_jk_diagonal_and_three_heavy_regime_60_synthesis`)

Unifies the two-window Cauchy tail-freeze (`SB.57`), the $j \geq k$ excess-exponent explosion and halving-debt barrier (`SB.58`), and the 3^{j_0} diagonal dead-zone invariance law (`SB.59`), completing all **60 `sorry` -free, warning-free theorems (`SB.1–SB.60`)**.

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15. Seed Finiteness, Extrema Seesaw, & The Trough-Rebound Trap (Theorems `SB.61–SB.64`)

In Sections 61–64 of `SemigroupBreakoutTheorems.lean` (verified computationally across 14,235 configurations in `sandbox_wave9_seed_finiteness_and_extrema.py`), we formalized how the finiteness of the seed P_0 interacts with the local and global extrema of the odd core u_r and rung height s_r:

15.1 Theorem SB.61 (`local_descent_rung_jump_and_multi_step_descent_explosion`)

1. **Single-Step Descent Rung Jump:** Whenever the odd core does not increase ($u_{r+1} \leq u_r$) across $2^{s_{r+1}} u_{r+1} = 3^{s_r} u_r + 1$, the binary flush must strictly dominate 3^{s_r}:

$$\boxed{u_{r+1} \leq u_r \implies 3^{s_r} < 2^{s_{r+1}}, \qquad s_r \geq 2 \implies s_{r+1} \geq 4, \qquad s_r \geq 4 \implies s_{r+1} \geq 7, \qquad s_r \geq 7 \implies s_{r+1} \geq 12.}$$

2. **Multi-Step Descent Staircase Explosion (`2 -> 4 -> 7 -> 12`):** Across any m-step non-increasing run ($\forall k < m,\ u_{r+k+1} \leq u_{r+k}$), $3^{s_{r+k}} < 2^{s_{r+k+1}}$ at every step; in particular, across any 3-step non-increasing run $u_{r+3} \leq u_{r+2} \leq u_{r+1} \leq u_r$ starting from $s_r \geq 2$, the rung heights are forced along the exponential staircase $s_{r+1} \geq 4,\ s_{r+2} \geq 7,\ s_{r+3} \geq 12$.

15.2 Theorem SB.62 (`trough_rebound_compound_interest_and_local_minimum_trap`)

Across any two consecutive block transitions $r \to r+1 \to r+2$:

$$\boxed{2^{s_{r+1}+s_{r+2}} u_{r+2} = 3^{s_r+s_{r+1}} u_r + 3^{s_{r+1}} + 2^{s_{r+1}}.}$$

Whenever step $r \to r+1$ enters a trough ($u_{r+1} \leq u_r$ with $s_r \geq 2$), the large binary flush $s_{r+1} \geq 4$ immediately becomes a $3^{s_{r+1}} \geq 81$ **compound-interest multiplier** on the rebound step $r+1 \to r+2$:

$$\boxed{81 u_{r+1} + 1 \leq 2^{s_{r+2}} u_{r+2} \qquad \text{and} \qquad 81\big(3^{s_r} u_r + 1\big) + 16 \leq 16 \cdot 2^{s_{r+2}} u_{r+2},}$$

and if s_{r+1} is a local rung peak ($2 \leq s_{r+2} \leq s_{r+1}$), then $9 \cdot 3^{s_{r+1} - s_{r+2}} u_{r+1} + 1 \leq 4 u_{r+2}$.

15.3 Theorem SB.63 (`finite_seed_global_peak_trough_seesaw_and_return_barrier`)

Across $K \geq 1$ blocks (T_K steps) starting from a finite seed $P_0 = 2^{s_0} u_0$ with $\forall k < K,\ 2 \leq s_k$: 1. **Strict Global Seesaw Sandwich on $2^{s_K} \cdot u_K$:**

$$\boxed{3^{T_K} \cdot 2^{s_0} u_0 < 2^{T_K + s_K} u_K \qquad \text{and} \qquad 5 \cdot 2^{T_K + s_K} u_K + 4 \cdot 2^{T_K} \leq 3^{T_K}\big(5 \cdot 2^{s_0} u_0 + 4\big).}$$

2. **Global Return-to-Baseline Rung Spike vs. Odd-Core Exponential Divergence:**

$$\boxed{u_K \leq u_0 \implies 3^{T_K} \cdot 2^{s_0} < 2^{T_K + s_K} \qquad \text{and} \qquad s_K \leq s_0 \implies 3^{T_K} u_0 < 2^{T_K} u_K}$$

3. **Post-Bit-Horizon Seed Exhaustion & Uniqueness ($P_0 < 2^{T_K+s_K}$):** $\lfloor 2^{T_K+s_K}u_K/3^{T_K}\rfloor = P_0$, $s_3(2^{T_K+s_K}u_K) = s_3(P_0)+s_3(A_{0,K})$, and whenever $P_0 < 2^{T_K+s_K}$, P_0 is the **unique** integer in $[0, 2^{T_K+s_K})$ satisfying $(3^{T_K}P_0' + A_{0,K}) \bmod 2^{T_K+s_K} = 0$.

15.4 Theorem SB.64 (`grand_seed_finiteness_and_extrema_64_synthesis`)

Unifies the local descent staircase explosion (`SB.61`), the trough-rebound $(81/16)$ compound-interest trap (`SB.62`), and the finite-seed global peak–trough seesaw (`SB.63`), completing all **64 `sorry` -free, warning-free theorems (`SB.1–SB.64`)**.

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16. LTE_2 Ultrametric Flush Pinning, Shifted-Basepoint Baker–Matveev Pinch, & The Senge–Straus Finiteness Bridge (Theorems `SB.65–SB.68`)

In Sections 65–68 of `SemigroupBreakoutTheorems.lean` (verified computationally across 6,040 configurations in `sandbox_wave10_lte2_baker_and_finiteness.py`), we formalized the exact 2-adic ultrametric flush-pinning table and bridged the multi-block identities directly to Baker's linear forms in logarithms and Senge–Straus / Stewart simultaneous (s_2, s_3) finiteness:

16.1 Theorem SB.65 (`lte2_ultrametric_flush_pinning_and_descent_prohibition`)

By the 2-adic ultrametric valuation law on $3^{s_r}u_r + 1 = (3^{s_r}-1)u_r + (u_r+1)$ (even s_r) and $3(3^{s_r-1}-1)u_r + (3u_r+1)$ (odd s_r), across all four residue classes of $s_r \bmod 4 \in \{0,1,2,3\}$ ($s_r, s_{r+1} \geq 2$):

$s_r \bmod 4$	$s_{r+1}=2$ **iff**	$s_{r+1}=3$ **(strict ascent $u_r<u_{r+1}$) iff**	**Descent-capable flush $4 \leq s_{r+1}$ iff**
2	$u_r \equiv 3 \pmod 8$	$u_r \equiv 15 \pmod{16}$	$u_r \equiv 7 \pmod{16}$
0	$u_r \equiv 3 \pmod 8$	$u_r \equiv 7 \pmod{16}$	$u_r \equiv 15 \pmod{16}$
3	$u_r \equiv 1 \pmod 8$	$u_r \equiv 5 \pmod{16}$	$u_r \equiv 13 \pmod{16}$
1	$u_r \equiv 1 \pmod 8$	$u_r \equiv 13 \pmod{16}$	$u_r \equiv 5 \pmod{16}$

In particular, when $s_r \equiv 2 \pmod 4$, having $16 \mid (u_r+1)$ ($u_r \equiv 15 \pmod{16}$) **pins $s_{r+1}=3$ and strictly forbids any descent flush $u_{r+1} \leq u_r$.**

16.2 Theorem SB.66 (`post_horizon_binary_zero_bit_desert_and_dual_radix_lock`)

Whenever the finite seed satisfies $P_0 < 2^{B_0}$ and $B_0 \leq S_K = T_K + s_K$: 1. **Binary Zero-Bit Desert of P_0 in $[B_0, S_K]$:** $\lfloor P_0/2^{B_0}\rfloor = 0$, $P_0 \bmod 2^{S_K} = P_0$, and for every $B \geq B_0$, $\lfloor (P_0 \bmod 2^{S_K})/2^B \rfloor = 0$. 2. **Simultaneous Base-2 and Base-3 Lock on $N_K = 2^{S_K}u_K = 3^{T_K}P_0 + A_{0,K}$:** $N_K \bmod 2^{S_K} = 0$, $\lfloor N_K/2^{S_K}\rfloor = u_K$, $s_2(N_K) = s_2(u_K)$, $\lfloor N_K/3^{T_K}\rfloor = P_0$, $N_K \bmod 3^{T_K} = A_{0,K}$, and $s_3(N_K) = s_3(P_0)+s_3(A_{0,K})$.

16.3 Theorem SB.67 (`shifted_basepoint_baker_matveev_pinch_and_repeated_core_rhin_form`)

Why does naive Baker fail at step 0? Because in $2^{T_K+s_K}u_K - 3^{T_K}P_0 = A_{0,K} \sim \frac{4}{5}3^{T_K}$, dividing by $3^{T_K}P_0$ leaves $\frac{4}{5P_0} = O(1)$, which does not decay with time. By **shifting the basepoint from step 0 to block $r \geq 1$** and advancing $M \geq 1$ blocks ($n_2 = T_{r+M} - T_r$ steps, $s_K = s_{r+M}$): 1. **Shifted-Basepoint 3^{n_2}-Canceled Baker–Matveev Pinch:**

$$\boxed{0 < 2^{n_2+s_K}u_{r+M} - 3^{n_2}\big(2^{s_r}u_r\big) = A_{r,M} \leq \frac{4}{5}\big(3^{n_2}-2^{n_2}\big), \qquad 3^{T_r}\cdot 2^{s_0}u_0 < 2^{T_r}\big(2^{s_r}u_r\big).}$$

Here the 3^{n_2} factor **cancels completely** when dividing by the main term $3^{n_2}(2^{s_r}u_r)$, leaving a relative linear form bounded by $\frac{4}{5\cdot 2^{s_r}u_r} < \frac{4}{5P_0}\left(\frac{2}{3}\right)^{T_r}$ that decays exponentially in T_r. 2. **Repeated-Odd-Core ($u_{r+M}=u_r$) Pure 2-Logarithm Rhin Reduction:** Whenever the odd core repeats at two blocks ($u_{r+M}=u_r$), the entire return integer $P_r = 2^{s_r}u_r$ factors out and **divides** $A_{r,M}$, leaving the pure 2-logarithm Rhin difference $2^{n_2+s_K-s_r} - 3^{n_2}$:

$$\boxed{\big(2^{s_r}u_r\big)\big(2^{n_2+s_K-s_r}-3^{n_2}\big) = A_{r,M}, \qquad 5\big(2^{s_r}u_r\big) + 4\cdot 2^{n_2} \leq 4\cdot 3^{n_2}, \qquad 5\cdot 3^{T_r}\cdot 2^{s_0}u_0 < 4\cdot 2^{T_r}\cdot 3^{n_2}}$$

16.4 Theorem SB.68 (`senge_straus_stewart_finiteness_bridge_and_grand_68_synthesis`)

Packages the exact Senge–Straus / Stewart simultaneous (s_2, s_3) finiteness bridge on $N_K = 2^{S_K}u_K = 3^{T_K}P_0 + A_{0,K}$:

$$\boxed{s_2(N_K) = s_2(u_K), \qquad s_3(N_K) = s_3(P_0)+s_3(A_{0,K}), \qquad s_3(N_K) + 2\big(v_3(N_K!) - 2^{S_K}v_3(u_K!)\big) = 2^{S_K}s_3(u_K),}$$

together with the conditional Senge–Straus closure (`s_2(u_K) ≤ B_2 → s_3(P_0) + s_3(A_{0,K}) ≤ B_3`), completing all **68 `sorry` -free, warning-free theorems (`SB.1–SB.68`)**.

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17. End-of-Block 3^{s_r}-Upgraded Baker Pinch, Adjacent/2-Step Return Extinction, & The $1/5$ Reduced Ceiling (Theorems `SB.69–SB.72`)

In Sections 69–72 of `SemigroupBreakoutTheorems.lean` (verified computationally across 9,920 configurations in `sandbox_wave11_upgraded_baker_and_return_extinction.py`), we formalized the exact 2-adic factorization of the renewal sum $A_{r,M} = 2^{s_r} R$ (R odd), upgraded the Baker pinch denominator from $2^{s_r} u_r$ to $3^{s_r} u_r$, proved unconditional non-equality $u_{r+1} \neq u_r$, and established the $1/5$ reduced renewal ceiling:

17.1 Theorem SB.69 (`adjacent_odd_core_coprimality_and_universal_non_equality`)

Across any single block transition $2^{s_{r+1}} u_{r+1} = 3^{s_r} u_r + 1$ with $s_r \geq 2$:

$$\boxed{\gcd(u_r, u_{r+1}) = 1, \qquad \gcd\left(2^{s_{r+1}} u_{r+1},\ 3^{s_r} u_r\right) = 1, \qquad u_{r+1} \neq u_r \text{ unconditionally}}$$

(Proof: $u_{r+1} = u_r$ *and* $\gcd(u_r, u_{r+1}) = 1$ *force* $u_r = 1$ *and* $2^{s_{r+1}} = 3^{s_r} + 1 \geq 10$*, so* $s_{r+1} \geq 4$ *and* $8 \mid 2^{s_{r+1}}$*, contradicting* $(3^{s_r} + 1) \bmod 8 \in \{2, 4\}$.

17.2 Theorem SB.70 (`two_step_odd_core_gcd_and_return_extinction`)

Across any two block transitions $r \to r+1 \to r+2$:

$$\boxed{\gcd(u_r, u_{r+2}) \,\Big|\, \left(3^{s_{r+1}} + 2^{s_{r+1}}\right), \qquad \text{and} \qquad u_{r+2} = u_r \implies u_r \,\Big|\, \left(3^{s_{r+1}} + 2^{s_{r+1}}\right) \ \wedge\ u_r \leq 3^{s_{r+1}} + 2^{s_{r+1}},}$$

completely **independent of** s_r. In addition, the unique boundary solution $(s_r, s_{r+1}, s_{r+2}, u_r, u_{r+1}) = (3, 2, 6, 1, 7)$ lands on $u_{r+2} = 1$ at even rung $s_{r+2} = 6$ and dies immediately at the next step ($(3^6 \cdot 1 + 1) \bmod 4 = 2 \implies s_{r+3} = 1$).

17.3 Theorem SB.71 (`end_of_block_three_pow_upgraded_baker_pinch_and_s_r_free_return_bound`)

Across $M+1$ blocks starting from u_r ($d = s_{r+1} + \cdots + s_{r+M}$, $S = d + s_{r+1+M}$, $T = s_r + d$, $R = 3^d + \text{renewalNumerator}(s', M)$): 1. **Strict Oddness & GCD Divisibility of R:** $2^S u_{r+1+M} = 3^T u_r + R$, $R \bmod 2 = 1$, and $\gcd(u_r, u_{r+1+M}) \mid R$. 2. **The 3^{s_r}-Upgraded Baker–Matveev Pinch:**

$$\boxed{5 \cdot 3^{s_r}\left(2^S u_{r+1+M} - 3^T u_r\right) + 4 \cdot 3^{s_r} \cdot 2^d \leq 9 \cdot 3^T, \qquad \text{so} \qquad 0 < \frac{2^S u_{r+1+M}}{3^T u_r} - 1 < \frac{9}{5 \cdot 3^{s_r} u_r}}$$

3. **s_r-Free M-Step Return Bound ($u_{r+1+M} = u_r$):**

$$\boxed{u_{r+1+M} = u_r \implies \left(2^S - 3^T\right) u_r = R, \qquad u_r \mid R, \qquad 5u_r + 4 \cdot 2^d \leq 9 \cdot 3^d,}$$

bounding u_r purely in terms of the intermediate step sum d, **completely independent of s_r and T_r.**

17.4 Theorem SB.72 (`exact_two_adic_renewal_valuation_and_one_fifth_reduced_ceiling`)

For any $K = M + 1 \geq 1$ blocks with $s_k \geq 2$, setting $d_0 = T_K - s_0$, $S_0 = d_0 + s_K$, and $R_0 = 3^{d_0} + \text{renewalNumerator}(s', M)$: 1. **Exact 2-Adic Valuation of the Renewal Sum ($v_2(A_{0,K}) = s_0$):**

$$\boxed{A_{0,K} = 2^{s_0} \cdot R_0 \qquad \text{with} \qquad R_0 \bmod 2 = 1}$$

2. **Reduced Evolution & The $1/5$ Reduced Renewal Ceiling:**

$$\boxed{2^{S_0} u_K = 3^{T_K} u_0 + R_0, \qquad 5R_0 + 4 \cdot 2^{d_0} \leq 3^{T_K}, \qquad 5R_0 < 3^{T_K} \quad \left(R_0 < \tfrac{1}{5} \cdot 3^{T_K}\right)}$$

3. **Reduced Base-3 Recovery & Zero-Carry Split:** $\lfloor 2^{S_0} u_K / 3^{T_K} \rfloor = u_0$, $(2^{S_0} u_K) \bmod 3^{T_K} = R_0$, and $s_3(2^{S_0} u_K) = s_3(u_0) + s_3(R_0)$.

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18. Cubic/Quintic Odd-Core Subordinacy, Global Multiplicative-to-Additive Product Formula, & The Ridout Cancellation Bridge (Theorems `SB.73–SB.76`)

In Sections 73–76 of `SemigroupBreakoutTheorems.lean` (verified computationally across 10,151 configurations in `sandbox_wave12_global_dominance_and_product_formula.py`), we formalized the global polynomial subordinacy of u_K to 2^{T_K}, the exact telescoped product formula across all K blocks, and the two-window Ridout/Subspace cancellation bridge:

18.1 Theorem SB.73 (`cubic_and_quintic_odd_core_subordinacy_to_two_pow`)

Across $K = M + 1 \geq 1$ blocks with $s_k \geq 2$ ($S_0 = T_K - s_0 + s_K$): 1. **Linear $1/5$ Envelope:** $5 \cdot 2^{S_0} u_K < 3^{T_K}(5u_0 + 1)$. 2. **Cubic $(3, 5)$ Global Dominance Law ($3^3 = 27 < 32 = 2^5 \implies u_K^3 \ll (2^{T_K})^2$):**

$$\boxed{\left(5 \cdot 2^{S_0} u_K\right)^3 < 2^{5T_K}(5u_0 + 1)^3 \qquad \text{and} \qquad 125 \cdot 2^{3s_K} u_K^3 < 2^{2T_K + 3s_0}(5u_0 + 1)^3}$$

3. **Quintic $(5,8)$ Global Dominance Law ($3^5 = 243 < 256 = 2^8 \implies u_K^5 \ll (2^{T_K})^3$):**

$$\boxed{\left(5 \cdot 2^{S_0} u_K\right)^5 < 2^{8T_K}(5u_0+1)^5 \qquad \text{and} \qquad 3125 \cdot 2^{5s_K} u_K^5 < 2^{3T_K+5s_0}(5u_0+1)^5}$$

Consequently, even in the fastest possible drifting regime, the odd core u_K accounts for $< 36.91\%$ of the logarithmic height of N_K, while the pure prime power 2^{S_0} accounts for $> 63.09\%$.

18.2 Theorem SB.74 (`global_multiplicative_to_additive_product_formula_bridge`)

Defining `oddCoreProd u K` $= \prod_{r=0}^{K-1} u_r$ and `stepNumProd s u K` $= \prod_{r=0}^{K-1}(3^{s_r}u_r+1)$:

$$\boxed{2^{T_K+s_K}u_K\prod_{r=0}^{K-1}u_r = \left(3^{T_K}P_0 + A_{0,K}\right)\prod_{r=0}^{K-1}u_r = P_0\prod_{r=0}^{K-1}\left(3^{s_r}u_r+1\right), \qquad \text{so} \qquad P_0 \,\Big|\, \left(A_{0,K}\prod_{r=0}^{K-1}u_r\right)}$$

18.3 Theorem SB.75 (`two_window_ridout_subspace_tail_cancellation_bridge`)

Across window $[r, r+M]$ of step sum $n_2 = T_{r+M} - T_r$ with $s_K = s_{r+M}$, the scaled tail $\Delta = 2^{T_r+n_2+s_K}u_{r+M} - 3^{n_2}(2^{T_r+s_r}u_r) = 2^{T_r}A_{r,M}$ satisfies:

$$\boxed{5\Delta + 4\cdot 2^{T_r+n_2} \le 4\cdot 2^{T_r}\cdot 3^{n_2} \qquad \text{and} \qquad 5\Delta < 4\cdot 2^{T_r}\cdot 3^{n_2}.}$$

Whenever a Ridout/Subspace lower bound $C_1 \cdot 2^{T_r+s_r}\cdot 3^{n_2} \le C_2 \cdot 3^E \cdot \Delta$ holds ($C_2 \ge 1$), **both 2^{T_r} and 3^{n_2} cancel completely**, forcing $5C_1 \cdot 2^{s_r} < 4C_2 \cdot 3^E$.

18.4 Theorem SB.76 (`grand_global_dominance_and_product_formula_76_synthesis`)

Unifies `SB.73–SB.75` with Wave 11 (`SB.69–SB.72`), completing all **76 `sorry`-free, warning-free theorems (`SB.1–SB.76`)**.

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19. Greedy 2-Adic Base-$(2/3)$ Kempner Expansion, Two-Horizon Kempner Tail Valuation, & The Rational-Target Obstruction Bridge (Theorems `SB.77–SB.80`)

In Sections 77–80 of `SemigroupBreakoutTheorems.lean` (verified computationally across 10,032 configurations in `sandbox_wave13_kempner_two_adic_expansion.py`), we formalized the exact equivalence between the Mahler block recurrence and the greedy 2-adic base-$(2/3)$ Kempner expansion of $-P_0$:

19.1 Theorem SB.77 (`greedy_two_adic_base_two_thirds_kempner_step_and_prefix_law`)

Because $|2/3|_\infty = 2/3 < 1$ and $|2/3|_2 = 1/2 < 1$, the Kempner series $\sum_{r=1}^{\infty}(2/3)^{T_r}$ converges simultaneously in $\mathbb{R}$ and in $\mathbb{Q}_2$. Across any `ValidBlockChain b_0 s u (K + 1)` with $s_k \ge 2$: 1. **Greedy 2-Adic Base-$(2/3)$ Prefix Valuation Law (`∀ r ≤ K`):**

$$\boxed{3^{T_r}P_0 + A_{0,r} = 2^{T_{r+1}}u_r, \qquad \left(3^{T_r}P_0 + A_{0,r}\right) \bmod 2^{T_{r+1}} = 0, \qquad \left(3^{T_r}P_0 + A_{0,r}\right) \bmod 2^{T_{r+1}+1} = 2^{T_{r+1}}}$$

2. **Greedy Single-Digit Base-$(2/3)$ Remainder Update (`∀ r < K`):** Adding the $(r+1)$-th base-$(2/3)$ digit $2^{T_{r+1}}$ updates the cleared partial sum via:

$$\boxed{3^{s_r}\left(3^{T_r}P_0 + A_{0,r}\right) + 2^{T_{r+1}} = 3^{T_{r+1}}P_0 + A_{0,r+1} = 2^{T_{r+1}}\left(3^{s_r}u_r+1\right) = 2^{T_{r+2}}u_{r+1}.}$$

19.2 Theorem SB.78 (`two_horizon_kempner_tail_exact_valuation_and_one_fifth_bound`)

Between any two horizons $K_1 = r$ and $K_2 = r+1+M$ ($s_r \ge 2$, $s_{r+1+k} \ge 2$, $T_{1,\text{next}} = T_r + s_r$, $d = \sum_{k=0}^{M-1} s_{r+1+k}$, $R_{\text{tail}} = 3^d + \text{renewalNumerator}(s', M)$): 1. **Exact Two-Horizon Kempner Tail Decomposition:**

$$\boxed{2^{T_{K_2}+s_{K_2}}u_{K_2} - 3^{s_r+d}\left(2^{T_{1,\text{next}}}u_{K_1}\right) = 2^{T_{1,\text{next}}}\cdot R_{\text{tail}} \qquad \text{with} \qquad R_{\text{tail}} \bmod 2 = 1,}$$

so the cleared Kempner tail between K_1 and K_2 has **exact 2-adic valuation T_{K_1+1}**. 2. **The $1/5$ Reduced Ceiling on the Kempner Tail Core:**

$$\boxed{5R_{\text{tail}} + 4\cdot 2^d \le 3^{s_r+d} \qquad \text{and} \qquad 5R_{\text{tail}} < 3^{s_r+d}}$$

19.3 Theorem SB.79 (`kempner_rational_target_obstruction_and_automaton_irrationality_bridge`)

Packages the simultaneous $\mathbb{R} \times \mathbb{Q}_2$ Kempner partial-sum lock with the 2-adic non-integrality exclusion law: for any candidate integer $M_0 \in \mathbb{N}$, if a 2-adic Mahler/Kempner transcendence or irrationality certificate establishes $\left(3^{T_K}M_0 + A_{0,K}\right) \bmod 2^{T_K+s_K} \neq 0$, then $P_0 \neq M_0$.

19.4 Theorem SB.80 (`grand_kempner_two_adic_expansion_80_synthesis`)

Unifies `SB.77–SB.79` with Wave 12 (`SB.73–SB.76`), completing all **80 `sorry`-free, warning-free theorems (`SB.1–SB.80`)**.

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20. The $(5P+4)/36$ Ceiling-Dual Gauge Transformation & Dual-Wall Monotone Squeeze (Theorems `SB.81–SB.84`)

In Sections 81–84 of `SemigroupBreakoutTheorems.lean` (verified computationally across 1,129 configurations in `sandbox_wave14_ceiling_dual_gauge_tower.py`), we introduced a new coordinate system outside the standard bottom-up renewal semigroup—the **Ceiling-Dual Gauge** $W_r := \frac{5P_r+4}{36}$, which measures distance **down** from the $4/5$ renewal ceiling rather than **up** from 0:

20.1 Theorem SB.81 (`ceiling_dual_gauge_transform_and_rung_two_parity_detector`)

For every interior block $r+1 \geq 1$ with $s_r, s_{r+1}, s_{r+2} \geq 2$, the return integer $P_{r+1} = 2^{s_{r+1}}u_{r+1} = 3^{s_r}u_r + 1$ satisfies $P_{r+1} \equiv 28 \pmod{36}$, so $36 \mid (5P_{r+1}+4)$ and $W_{r+1} = (5P_{r+1}+4)/36 \in \mathbb{N}$ is an exact positive integer satisfying: 1. **Rung-2 Parity Detector:**

$$\boxed{9W_{r+1} = 5 \cdot 2^{s_{r+1}-2}u_{r+1} + 1, \qquad W_{r+1} \bmod 2 = 0 \iff s_{r+1} = 2, \qquad W_{r+1} \bmod 2 = 1 \iff s_{r+1} \geq 3}$$

2. **One-Step Sub-Multiplicative Gauge Law & Pure $(9/4)$ Step at $s_{r+1} = 2$:**

$$\boxed{2^{s_{r+1}}W_{r+2} + \left(3^{s_{r+1}-2} - 2^{s_{r+1}-2}\right) = 3^{s_{r+1}}W_{r+1}, \qquad \text{and} \qquad s_{r+1} = 2 \implies 4W_{r+2} = 9W_{r+1}}$$

Every $s = 2$ step has **zero additive error** in the W-gauge and consumes exactly 2 powers of 2 from $v_2(W)$, so $v_2(W_{r+1})$ is the exact countdown clock for consecutive $s = 2$ steps.

20.2 Theorem SB.82 (`multi_block_gauge_deficit_master_identity_and_one_ninth_bound`)

Across any $M \geq 1$ blocks starting from $r+1$ ($d = \sum_{k=0}^{M-1} s_{r+1+k}$), defining `gaugeDeficit s' M` ($E_M$) recursively:

$$\boxed{5A_M + 4 \cdot 2^d + 36E_M = 4 \cdot 3^d, \qquad 2^d W_{r+1+M} + E_M = 3^d W_{r+1}, \qquad 9E_M + 2^d \leq 3^d \quad \left(E_M < 3^{d-2}\right)}$$

20.3 Theorem SB.83 (`gauge_complement_base3_conservation_and_top_22_trit_lock`)

Whenever $E_M \geq 1$ (at least one block in the window has $s_{r+1+k} \geq 3$): 1. **Exact Base-3 Quotient & Complement Remainder:**

$$\left\lfloor \frac{2^d W_{r+1+M}}{3^d} \right\rfloor = W_{r+1} - 1 \qquad \text{and} \qquad \left(2^d W_{r+1+M}\right) \bmod 3^d = 3^d - E_M.$$

2. **Top Two Trits Locked to `22` in Base 3 (`8 = 2*3 + 2`):**

$$\boxed{8 \cdot 3^{d-2} < 3^d - E_M < 3^d \qquad \text{and} \qquad \left\lfloor \frac{\left(2^d W_{r+1+M}\right) \bmod 3^d}{3^{d-2}} \right\rfloor = 8}$$

3. **Exact Gauge-Complement Base-3 Conservation Law:**

$$\boxed{s_3\left(2^d W_{r+1+M}\right) + s_3(E_M - 1) = s_3(W_{r+1} - 1) + 2d}$$

20.4 Theorem SB.84 (`dual_wall_monotone_squeeze_and_grand_84_synthesis`)

Traps every interior window $[r+1, r+1+M]$ simultaneously between the **strictly super-multiplicative Lower-Wall sequence** P_k ($3^d P_{r+1} < 2^d P_{r+1+M}$) and the **sub-multiplicative Upper-Wall Ceiling-Dual Gauge sequence** W_k ($2^d W_{r+1+M} \leq 3^d W_{r+1}$) coupled by $36W_k = 5P_k + 4$, completing all **84 `sorry`-free, warning-free theorems (`SB.1–SB.84`)**.

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21. Second-Order Super-Block Renormalization, Mod-8 Future-Rung Predictor, & The Cyclotomic Gauge Spectrum (Theorems `SB.85–SB.88`)

In Sections 85–88 of `SemigroupBreakoutTheorems.lean` (verified computationally across 7,772 configurations in `sandbox_wave15_superblock_and_multigauge.py`), we formalized the second-order "Super-Block" renormalization $W_{r+1} = 4^m V$ that collapses entire runs of $s = 2$ blocks into a single macro-step, predicts the first non-2 rung m blocks in advance, and embeds W_r into the infinite Cyclotomic Gauge Spectrum $\mathcal{G}_r^{(m)} := (3^m - 2^m)P_r + 2^m$:

21.1 Theorem SB.85 (`gauge_odd_valuation_trap_and_mod_four_survival_lock`)

1. **Delayed-Action Death Trap #1 ($W_{r+1} \bmod 4 \neq 2$):** In any surviving chain with $s_r, s_{r+1}, s_{r+2} \geq 2$:

$$\boxed{W_{r+1} \bmod 4 \neq 2, \qquad \left(W_{r+1} \bmod 2 = 0 \iff W_{r+1} \bmod 4 = 0\right), \qquad \left(s_{r+1} = 2 \iff W_{r+1} \bmod 4 = 0\right)}$$

2. **Multi-Step $s = 2$ Run Evolution & Odd-Valuation Extinction:** Across any run of m consecutive $s = 2$ blocks ($s_{r+1+k} = 2$ for $k < m$), $4^m W_{r+1+m} = 9^m W_{r+1}$, and for every odd V, $W_{r+1} \neq 2 \cdot 4^m \cdot V$ ($v_2(W_{r+1})$ can never be odd).

21.2 Theorem SB.86 (`mod_eight_odd_gauge_future_rung_predictor_and_seven_extinction`)

Suppose $W_{r+1} = 4^m \cdot V$ executes m consecutive steps of $s = 2$ followed by the first non-2 rung $s^* = s_{r+1+m} \geq 3$ (with $s_{r+1+m+1} \geq 2$): 1. **Exact Super-Block Arrival & Mod-8 Invariance:**

$$\boxed{W_{r+1+m} = 9^m \cdot V, \qquad V \bmod 2 = 1, \qquad W_{r+1+m} \bmod 8 = V \bmod 8}$$

2. **Delayed-Action Death Trap #2 ($V \bmod 8 \neq 7$) & Future-Rung Predictor m Blocks in Advance:**

$$\boxed{V \bmod 8 \neq 7, \qquad V \bmod 8 = 3 \iff s^* = 3, \qquad V \bmod 8 = 5 \iff s^* = 4, \qquad V \bmod 8 = 1 \iff s^* \geq 5}$$

3. **Exact Super-Block Jump to the Next Gauge State:**

$$\boxed{2^{s^*} W_{r+1+m+1} + \left(3^{s^*-2} - 2^{s^*-2}\right) = 3^{2m+s^*} \cdot V}$$

21.3 Theorem SB.87 (`universal_cyclotomic_gauge_spectrum_and_two_three_eliminator`)

For every integer $m \geq 1$, the m-th Cyclotomic Gauge $\mathcal{G}_r^{(m)} := (3^m - 2^m)P_r + 2^m$ satisfies:

$$\boxed{\begin{aligned} &s_r = m \implies 2^m \mathcal{G}_{r+1}^{(m)} = 3^m \mathcal{G}_r^{(m)} \quad (\text{exact annihilation at } s_r = m), \\ &m \leq s_r \implies 2^{s_r} \mathcal{G}_{r+1}^{(m)} + 2^m 3^m \left(3^{s_r - m} - 2^{s_r - m}\right) = 3^{s_r} \mathcal{G}_r^{(m)}, \\ &s_r \leq m \implies 2^{s_r} \mathcal{G}_{r+1}^{(m)} = 3^{s_r} \mathcal{G}_r^{(m)} + 2^{s_r} 3^{s_r} \left(3^{m - s_r} - 2^{m - s_r}\right), \end{aligned}}$$

and the $m = 2$ ($5P_r + 4$) and $m = 3$ ($19P_r + 8$) gauges satisfy the exact linear eliminator $19\mathcal{G}_r^{(2)} = 5\mathcal{G}_r^{(3)} + 36$.

21.4 Theorem SB.88 (`grand_superblock_and_multigauge_88_synthesis`)

Unifies `SB.85–SB.87` with Wave 14 (`SB.81–SB.84`), completing all **88** `sorry`**-free, warning-free theorems (`SB.1–SB.88`)**.

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22. Adaptive Cyclotomic Velocity Calculus, Shrinking Canyon Tube, & Master Conditional Closure (Theorems `SB.89–SB.92`)

In Sections 89–92 of `SemigroupBreakoutTheorems.lean` (verified computationally across 6,546 configurations in `sandbox_wave16_velocity_calculus_and_master_closure.py`), we formalized the **Adaptive Cyclotomic Velocity Calculus** ($K_r := P_{r+1} - P_r = (3^{s_r} - 2^{s_r})u_r + 1$), the **Exponentially Shrinking Nested Canyon Tube**, and the **Master Conditional Completeness Closure**:

22.1 Theorem SB.89 (`adaptive_cyclotomic_velocity_factorization_and_eliminator`)

Defining the Orbit Velocity $K_r := (3^{s_r} - 2^{s_r})u_r + 1$: 1. **Exact First-Difference Velocity Identity:**

$$P_r + K_r = P_{r+1}, \qquad P_r < P_{r+1}, \qquad P_{r+1} - P_r = K_r.$$

2. **Adaptive Cyclotomic Gauge Factorization Through K_r ($m = s_r$):**

$$\boxed{\mathcal{G}^{(s_r)}(P_r) = 2^{s_r} \cdot K_r \qquad \text{and} \qquad \mathcal{G}^{(s_r)}(P_{r+1}) = 3^{s_r} \cdot K_r}$$

3. **Equal-Rung Velocity Countdown Law:** Whenever $s_{r+1} = s_r$, $2^{s_r} K_{r+1} = 3^{s_r} K_r$.

22.2 Theorem SB.90 (`velocity_two_adic_valuation_lock_and_telescoped_odd_core_sum`)

1. **Exact 2-Adic Valuation Lock at Every Rung Change ($s_{r+1} \neq s_r$):**

$$\boxed{\begin{aligned} &s_r < s_{r+1} \implies K_r = 2^{s_r}\left(2^{s_{r+1} - s_r} u_{r+1} - u_r\right) \ \wedge \ K_r \bmod 2^{s_r+1} = 2^{s_r} \quad \left(v_2(K_r) = s_r\right), \\ &s_{r+1} < s_r \implies K_r = 2^{s_{r+1}}\left(u_{r+1} - 2^{s_r - s_{r+1}} u_r\right) \ \wedge \ K_r \bmod 2^{s_{r+1}+1} = 2^{s_{r+1}} \quad \left(v_2(K_r) = s_{r+1}\right) \end{aligned}}$$

2. **Telescoped Velocity Sum Across K Blocks:**

$$P_K = P_0 + \sum_{r=0}^{K-1} K_r \qquad \text{and} \qquad 2^{s_K} u_K = 2^{s_0} u_0 + \sum_{r=0}^{K-1} \left(\left(3^{s_r} - 2^{s_r}\right)u_r + 1\right).$$

22.3 Theorem SB.91 (`nested_canyon_interval_tube_and_exponential_width_contraction`)

Setting $L_{\text{num}}(k) := 5 \cdot 2^{T_k} P_k$ and $U_{\text{num}}(k) := 2^{T_k}(5P_k + 4)$ across any window $[r, r + M]$ of step sum d ($M \geq 1$, $s_{r+k} \geq 2$):

$$\boxed{\begin{aligned} U_{\mathrm{num}}(k) - L_{\mathrm{num}}(k) &= 4 \cdot 2^{T_k} \quad \left(\text{canyon width } U_k - L_k = 4 \cdot (2/3)^{T_k}\right), \\ 3^d L_{\mathrm{num}}(r) + 5 \cdot 2^{T_r} A_M &= L_{\mathrm{num}}(r+M), \\ U_{\mathrm{num}}(r+M) + 36 \cdot 2^{T_r} E_M &= 3^d U_{\mathrm{num}}(r), \\ 3^d L_{\mathrm{num}}(r) < L_{\mathrm{num}}(r+M) &< U_{\mathrm{num}}(r+M) \le 3^d U_{\mathrm{num}}(r) \end{aligned}}$$

22.4 Theorem SB.92 (`master_conditional_canyon_closure_and_92_theorem_synthesis`)

Formalizes the exact conditional closure bridges for **Path A / Path C** (Carry-Mixing / Kempner Non-Alignment Obstruction: $(\forall P_0 < 2^{B_0},\ \forall u_K \text{ odd},\ 3^{T_K} P_0 + A_{0,K} \neq 2^{T_K+s_K} u_K) \implies \neg\mathrm{ValidBlockChain}\ b_0\ s\ u\ (K+1)$) and **Path B** (Multi-Step Product-Formula Obstruction: $\neg(P_0 \mid A_{0,K} \prod_{r=0}^{K-1} u_r) \implies \neg\mathrm{ValidBlockChain}\ b_0\ s\ u\ (K+1)$), completing all **92** `sorry`**-free, warning-free theorems (`SB.1–SB.92`)**.

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23. Path A (Binary Carry-Split & Quotient Decomposition of u_K) and Path B (Reduced Odd-Seed Product Formula & Symmetric Excess) (Theorems `SB.93–SB.96`)

In Sections 93–96 of `SemigroupBreakoutTheorems.lean` (verified computationally across 2,343 configurations in `sandbox_wave17_path_a_and_path_b.py`), we pursued both Path A and Path B to their exact structural decompositions:

23.1 Theorem SB.93 (`path_a_binary_carry_split_and_quotient_decomposition_of_u_K`)

In the reduced odd-seed evolution equation $2^{S_0} u_K = 3^{T_K} u_0 + R_0$ ($S_0 = T_K - s_0 + s_K$, $R_0 \bmod 2 = 1$): 1. **Exact 2^{S_0} Low-Bit 2's-Complement Cancellation:**

$$\boxed{1 \le \left(3^{T_K} u_0\right) \bmod 2^{S_0}, \qquad 1 \le R_0 \bmod 2^{S_0}, \qquad \left(3^{T_K} u_0\right) \bmod 2^{S_0} + \left(R_0 \bmod 2^{S_0}\right) = 2^{S_0}}$$

2. **Exact Binary Quotient Decomposition of the Odd Core u_K:**

$$\boxed{u_K = \left\lfloor \frac{3^{T_K} u_0}{2^{S_0}} \right\rfloor + \left\lfloor \frac{R_0}{2^{S_0}} \right\rfloor + 1}$$

3. **Exact Carry-Mixing Binary Weight Conservation Law:**

$$\boxed{s_2\Big(\left(3^{T_K} u_0\right) \bmod 2^{S_0}\Big) + s_2\Big(R_0 \bmod 2^{S_0}\Big) = S_0 + 1}$$

23.2 Theorem SB.94 (`path_a_zero_carry_binary_block_split_and_weight_excess_law`)

Splitting $3^{T_K} u_0$ and R_0 in Base 2 at bit S_0 yields the exact total binary weight identity across the carry boundary:

$$\boxed{s_2\left(3^{T_K} u_0\right) + s_2(R_0) = S_0 + 1 + s_2\left(\left\lfloor \frac{3^{T_K} u_0}{2^{S_0}} \right\rfloor\right) + s_2\left(\left\lfloor \frac{R_0}{2^{S_0}} \right\rfloor\right),}$$

forcing $s_2(3^{T_K} u_0) + s_2(R_0) \ge S_0 + 1$, and $s_2(3^{T_K} u_0) + s_2(R_0) \ge S_0 + 2$ whenever $u_K \ge 2$.

23.3 Theorem SB.95 (`path_b_reduced_odd_seed_product_formula_and_symmetric_excess`)

Canceling the trivial seed factor $P_0 = 2^{s_0} u_0$ from the global product formula (`SB.74`) with `tailOddProd` $= \prod_{r=1}^{K-1} u_r$:

$$\boxed{\begin{aligned} \left(3^{T_K} u_0 + R_0\right) \prod_{r=1}^{K-1} u_r &= \prod_{r=0}^{K-1} \left(3^{s_r} u_r + 1\right), \\ R_0 \prod_{r=1}^{K-1} u_r &= \prod_{r=0}^{K-1} \left(3^{s_r} u_r + 1\right) - 3^{T_K} \prod_{r=0}^{K-1} u_r, \\ \prod_{r=0}^{K-1} \left(3^{s_r} u_r + 1\right) \bmod u_0 &= \left(R_0 \prod_{r=1}^{K-1} u_r\right) \bmod u_0 \end{aligned}}$$

23.4 Theorem SB.96 (`grand_path_a_and_path_b_96_theorem_synthesis`)

Unifies Path A (`SB.93–SB.94`) and Path B (`SB.95`) with the Master Conditional Closure (`SB.92`), completing all **96** `sorry`**-free, warning-free theorems (`SB.1–SB.96`)**.

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24. Universal Mod-5/Mod-10 Gauge Lock, Mod-40 Six-Residue Super-Block Bijection, & Cyclotomic Base-D_m Tower (Theorems `SB.97–SB.100`)

In Sections 97–100 of `SemigroupBreakoutTheorems.lean` (verified computationally across 8,114 configurations in `sandbox_wave18_mod40_and_multibase_tower.py`), we formalized the multi-base modular locks across Base 5, Base 10, Base 40, Base 4, Base 9, and the Cyclotomic Bases $D_m = 3^m - 2^m$, completing the **Centennial 100-Theorem Semigroup Breakout Library (`SB.1–SB.100`)**:

24.1 Theorem SB.97 (`universal_mod5_and_mod10_gauge_lock_and_run_parity_detector`)

1. **Universal Mod-5 & Mod-10 Gauge Lock on W_{r+1}:** Because $36W_{r+1} = 5P_{r+1} + 4$ and $36 \equiv 1 \pmod 5$, at every interior block $r+1$ ($s_r, s_{r+1}, s_{r+2} \geq 2$):
$$\boxed{W_{r+1} \bmod 5 = 4, \qquad \big(W_{r+1} \bmod 10 = 4 \iff s_{r+1} = 2\big), \qquad \big(W_{r+1} \bmod 10 = 9 \iff 3 \leq s_{r+1}\big)}$$
2. **Mod-5 & Mod-10 Super-Block Run-Length Parity Detector:** Across any Super-Block $W_{r+1} = 4^m \cdot V$ ending at $s^* = s_{r+1+m} \geq 3$:
$$\boxed{\big(m \bmod 2 = 0 \iff V \bmod 5 = 4 \iff V \bmod 10 = 9\big), \qquad \big(m \bmod 2 = 1 \iff V \bmod 5 = 1 \iff V \bmod 10 = 1\big),}$$
and $V \bmod 5 \notin \{0, 2, 3\}$.

24.2 Theorem SB.98 (`mod40_six_residue_superblock_bijection_and_34_class_extinction`)

Combining Mod-8 (`SB.86`) and Mod-5 (`SB.97`) via the Chinese Remainder Theorem locks every surviving Super-Block odd core V into an **exact 6-way bijection modulo** 40 (rendering **34 of 40 residue classes modulo 40 unconditionally extinct**):

$V \bmod 40$	$s = 2$ Run-Length Parity ($m \bmod 2$)	Terminating Rung $s^* = s_{r+1+m}$
19	$m \bmod 2 = 0$ (even)	$s^* = 3$
11	$m \bmod 2 = 1$ (odd)	$s^* = 3$
29	$m \bmod 2 = 0$ (even)	$s^* = 4$
21	$m \bmod 2 = 1$ (odd)	$s^* = 4$
9	$m \bmod 2 = 0$ (even)	$s^* \geq 5$
1	$m \bmod 2 = 1$ (odd)	$s^* \geq 5$

$$\boxed{V \bmod 40 \in \{1, 9, 11, 19, 21, 29\}}$$

24.3 Theorem SB.99 (`cyclotomic_base_dm_modular_invariance_and_mod19_mod13_locks`)

1. **Universal Base-D_m ($D_m = 3^m - 2^m$) Modular Lock:** For every $m, P \in \mathbb{N}$, $\mathcal{G}^{(m)}(P) \bmod (3^m - 2^m) = 2^m \bmod (3^m - 2^m)$; in particular:
$$\boxed{\mathcal{G}^{(2)}(P) \bmod 5 = 4, \quad \mathcal{G}^{(3)}(P) \bmod 19 = 8, \quad \mathcal{G}^{(4)}(P) \bmod 65 = 16, \quad \mathcal{G}^{(4)}(P) \bmod 13 = 3, \quad \mathcal{G}^{(5)}(P) \bmod 211 = 32}$$
2. **Base-4 to Base-9 Super-Block Shift & Base-3 Digit-Sum Invariance:** For $W_{\text{start}} = 4^m V$ and $W_{\text{end}} = 9^m V$, $\lfloor W_{\text{start}}/4^m \rfloor = V = \lfloor W_{\text{end}}/9^m \rfloor$, $W_{\text{start}} \bmod 4^m = 0 = W_{\text{end}} \bmod 9^m$, and $s_3(9^m V) = s_3(V)$.

24.4 Theorem SB.100 (`grand_centennial_100_theorem_semigroup_breakout_synthesis`)

The **Centennial 100-Theorem Grand Synthesis**, unifying all 100 `sorry`-free, warning-free theorems (`SB.1–SB.100`).

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25. Unconditional Density Decay, Fast $O(1)$ Super-Block Sieve, & The ZFC Deductive Semigroup Bridge (Theorems `SB.101–SB.104`)

In Sections 101–104 of `SemigroupBreakoutTheorems.lean` (verified computationally across 500,000 odd seeds $b_0 < 1{,}000{,}000$ in `sandbox_wave19_superblock_sieve_and_density.py`), we executed Paths 1–3: proving unconditional block-extinction and Mod-40 window-density bounds, deploying the $O(1)$ Super-Block Sieve to discover record-holding survivor seeds, and formalizing the pre-horizon vs. post-horizon ZFC deductive semigroup bridge:

25.1 Theorem SB.101 (`odd_seed_block_zero_and_block_one_exact_extinction_classes`)

Explains why **exactly $(1/2)^K$ of odd seeds b_0 survive K blocks**: 1. **Block-0 50% Extinction Class ($b_0 \equiv 3 \pmod 4$):** For every odd b_0, $4 \mid (3b_0 + 1) \iff b_0 \equiv 1 \pmod 4$, while $b_0 \equiv 3 \pmod 4 \implies (3b_0 + 1) \bmod 4 = 2$ ($s_0 = 1$, instant death at Block 0 for 50% of all odd seeds). 2. **Block-1 Unconditional Extinction Classes ($b_0 \equiv 1 \pmod{16}$, $b_0 \equiv 29 \pmod{32}$, $b_0 \equiv 5 \pmod{64}$):**

$$\boxed{\begin{aligned} b_0 \equiv 1 \pmod{16} &\implies s_0 = 2 \ \wedge\ u_0 \equiv 1 \pmod 4 \implies s_1 = 1 \quad \text{(dies at Block 1)},\\ b_0 \equiv 29 \pmod{32} &\implies s_0 = 3 \ \wedge\ u_0 \equiv 3 \pmod 4 \implies s_1 = 1 \quad \text{(dies at Block 1)},\\ b_0 \equiv 5 \pmod{64} &\implies s_0 = 4 \ \wedge\ u_0 \equiv 1 \pmod 4 \implies s_1 = 1 \quad \text{(dies at Block 1)} \end{aligned}}$$

Block Horizon K	Surviving Odd Seeds ($b_0 < 10^6$, out of $500{,}000$)	Empirical Survival Density	Theoretical $(1/2)^K$ Decay
$K = 1$	250,000	0.500000	0.500000
$K = 2$	125,000	0.250000	0.250000
$K = 3$	62,506	0.125012	0.125000
$K = 5$	15,630	0.031260	0.031250
$K = 8$	1,961	0.003922	0.003906
$K = 10$	482	0.000964	0.000977
$K = 15$	14	0.000028	0.000031
$K = 18$	1 ($b_0 = 415{,}789$)	0.000002	0.000004
$K = 20$	0	0.000000	0.000001

25.2 Theorem SB.102 (`mod40_window_thirty_four_of_forty_extinction_and_dyadic_uniqueness`)

Certifies that out of every 40 consecutive residues $0 \ldots 39$, **exactly 6 (15%) satisfy `isMod40Survivor` and 34 (85%) are extinct**, and that past the bit-horizon $P_0 < 2^{T_K + s_K}$, at most **one** seed in $[0, 2^{T_K+s_K})$ can realize a given rung sequence $(s_0, \ldots, s_{K-1})$.

25.3 Theorem SB.103 (`fast_superblock_sieve_macro_step_and_record_survivor_certificates`)

Proves the closed-form $O(1)$ Super-Block Sieve jump formula:

$$\boxed{W_{r+1} = 4^m \cdot V \implies P_{r+1+m} = \frac{36 \cdot 9^m \cdot V - 4}{5},}$$

which skips m consecutive $s = 2$ blocks in a single arithmetic operation, and certifies the record-holding survivor seeds discovered by the sieve:

Champion Seed b_0	Blocks Survived (K)	Total Mahler Steps (T_K)	Rung Sequence $[s_0, s_1, \ldots, s_K]$
13	7	21	$[3, 3, 2, 2, 2, 7, 2, 1]$
53	5	23	$[5, 6, 2, 6, 4, 1]$
149	5	27	$[6, 4, 4, 8, 5, 1]$
485	9	25	$[4, 2, 2, 2, 2, 2, 6, 2, 3, 1]$ *(skips five $s = 2$ blocks via $4^5 \to 9^5$)*
1,257	10	32	$[2, 3, 3, 4, 2, 2, 6, 3, 5, 2, 1]$
2,725	13	38	$[4, 4, 2, 2, 3, 4, 2, 2, 3, 4, 4, 2, 2, 1]$
51,877	15	47	$[4, 4, 2, 2, 2, 2, 2, 4, 3, 7, 4, 2, 4, 2, 3, 1]$
365,837	17	59	$[3, 3, 3, 6, 2, 3, 2, 2, 3, 2, 2, 2, 7, 7, 5, 4, 3, 1]$
415,789	18	46	$[3, 2, 2, 2, 2, 2, 3, 2, 4, 3, 2, 2, 4, 2, 2, 3, 3, 3, 1]$

25.4 Theorem SB.104 (`zfc_deductive_semigroup_pre_vs_post_horizon_overspill_bridge`)

Formalizes the exact bridge between the **pre-horizon 2-adic periodic lattice** ($\forall q \in \mathbb{N},\ P_0 + q \cdot 2^{S_K+1}$ satisfies the exact same K-block survival congruence) and the **post-horizon archimedean collapse** ($P_0 < 2^{S_K}$ excludes all $q \geq 1$, leaving at most one natural number).

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26. Generalized $q/2$ Relaxation Phase Diagram, Kempner Threshold Inversion ($q = 3$ vs. $q \geq 5$), & The $(q^2-4)u+1$ Core Gauge (`SB.105–SB.108`)

To isolate what distinguishes $3/2$ from other half-integers, we analyze the generalized confinement problem $\{x(q/2)^n\} \in [0, k)$ across odd numerators $q \in \{3, 5, 7, 9, 11, 13, 15, 17, 19, 21\}$. This comparison separates mechanisms shared by **every odd $q \geq 3$** from the Archimedean constraint that **uniquely over-determines $q = 3$ ($3/2$)**.

26.1 Theorem SB.105 (`universal_q_half_zone_thresholds_and_kempner_inversion_dichotomy`)

For any odd $q \geq 3$, the four critical relaxation thresholds for $\{x(q/2)^n\} \in [0, k)$ are: - $k_0(q) = \frac{1}{q}$ (Zone 0: total extinction, 0 odd steps), - $k_1(q) = \frac{q+2}{q^2}$ (Zone 1 top: $s_r = 1$ forbidden, $s_r \geq 2$ canyon), - $k_{\text{Kemp}}(q) = \frac{q}{q^2-4}$ (the all-$s_r = 2$ periodic Kempner fixed point), and - $k_3(q) = \frac{1}{q-2}$ (Zone 3: 100% survival, gapless coverage).

We prove the **Kempner Threshold Inversion Dichotomy** via the exact polynomial identity:

$$\boxed{(q+2)(q^2-4)-q^3 \;=\; 2\big(q^2-2q-4\big)\,:}$$

1. **At $q=3$ $(3/2)$:** $(3+2)(3^2-4)=25<27=3^3$, so $\frac{5}{9}<\frac{3}{5}$ $(k_1(3)<k_{\mathrm{Kemp}}(3))$, and at Mahler's $k=\frac{1}{2}$, $\frac{1}{2}<\frac{3}{5}$ with Kempner Overflow Ratio $\frac{\Phi_{\max}(3)}{3(1/2)-1}=\frac{4/5}{1/2}=\frac{8}{5}=\mathbf{1.600}>\mathbf{1}$. 2. **For every $q=d+4\geq 4$ (including every odd $q\geq 5$, i.e., $5/2, 7/2, 9/2, \ldots$):**

$$(q+2)(q^2-4) \;=\; q^3+2\big(d^2+6d+4\big) \;\geq\; q^3+8 \;>\; q^3 \;\Longrightarrow\; \boxed{\frac{q}{q^2-4} \;<\; \frac{q+2}{q^2}.}$$

Thus for every odd $q\geq 5$, the all-$s=2$ Kempner sum fits strictly inside Zone 1 ($\Phi_{\max}(q)/(2/q)=\frac{2q}{q^2-4}<0.48$), making $q=3$ **the sole odd fraction where Zone 1 suffers an Archimedean Kempner overflow on top of 2-adic mortality**.

Fraction $q/2$	Zone 0 ($k_0=\frac{1}{q}$)	Zone 1 Top ($k_1=\frac{q+2}{q^2}$)	Kempner Point ($k_{\mathrm{Kemp}}=\frac{q}{q^2-4}$)	Zone 3 ($k_3=\frac{1}{q-2}$)	Kempner Overflow Ratio $\frac{\Phi_{\max}}{\text{Budget}}$
$3/2$ ($q=3$)	$1/3\approx 0.3333$	$5/9\approx 0.5556$	$3/5=0.6000$ ($>k_1$)	1.0000	1.6000 at $k=1/2$ (1.2000 at k_1)
$5/2$ ($q=5$)	$1/5=0.2000$	$7/25=0.2800$	$5/21\approx 0.2381$ ($<k_1$)	$1/3\approx 0.3333$	$0.4762<1$
$7/2$ ($q=7$)	$1/7\approx 0.1429$	$9/49\approx 0.1837$	$7/45\approx 0.1556$ ($<k_1$)	$1/5=0.2000$	$0.3111<1$
$9/2$ ($q=9$)	$1/9\approx 0.1111$	$11/81\approx 0.1358$	$9/77\approx 0.1169$ ($<k_1$)	$1/7\approx 0.1429$	$0.2338<1$
$11/2$ ($q=11$)	$1/11\approx 0.0909$	$13/121\approx 0.1074$	$11/117\approx 0.0940$ ($<k_1$)	$1/9\approx 0.1111$	$0.1880<1$

26.2 Theorem SB.106 (`universal_q_half_mod4_one_step_extinction_and_four_adic_bijection`)

Establishes the exact **one-step** modular split and renewal uniqueness underlying the empirical $(1/2)^K$ Zone-1 survival decay observed across $100{,}000$ odd seeds: 1. **Even-Power Mod-8 Lock:** $((2m+1)^{2k}) \bmod 8=1$ and $((2m+1)^{2k+1}) \bmod 8=(2m+1) \bmod 8$. 2. **Universal Mod-4 50/50 One-Step Split:** For any odd $Q=q^{s_r}$, among every consecutive pair of odd cores $\{4t+1, 4t+3\}$, **exactly one** satisfies $(Qu+1) \bmod 4=0$ ($s_{r+1}\geq 2$, survives the step) and **exactly one** satisfies $(Qu+1) \bmod 4=2$ ($s_{r+1}=1$, dies). 3. **Universal Mod-2^{S_0} Renewal Uniqueness:** Because $\gcd((2m+1)^{T_K}, 2^{S_0})=1$, at most one seed $P_0<2^{S_0}$ can satisfy $((2m+1)^{T_K}P_0+A_K) \bmod 2^{S_0}=0$.

(Empirically across $100{,}000$ odd seeds in `sandbox_wave20_universal_q_half_relaxation_and_kempner_dichotomy.py`, iterating this 50/50 split produces sample survival fractions near $(1/2)^K$ and sample mean lifetimes in $[0.99968, 1.00073]$, consistent with an idealized independent geometric model of mean 1.)

Fraction $q/2$	$q \bmod 8$	Survives $K=1$	Survives $K=2$	Survives $K=5$	Max Blocks (10^5 seeds)	Sample Mean Blocks Survived
$3/2$	3	0.50000	0.25000	0.03132	15	1.00073
$5/2$	5	0.50000	0.25001	0.03137	18	0.99981
$7/2$	7	0.50000	0.25001	0.03128	17	0.99989
$9/2$	1	0.50000	0.24998	0.03117	21	0.99968
$11/2$	3	0.50000	0.25001	0.03143	18	0.99997
$13/2$	5	0.50000	0.25001	0.03110	14	1.00014
$15/2$	7	0.50000	0.24999	0.03138	15	0.99986

26.3 Theorem SB.107 (`universal_q_half_ceiling_dual_gauge_and_superblock_law`) & Theorem SB.108 (`grand_universal_q_half_relaxation_and_mahler_uniqueness_108_synthesis`)

Generalizes the Ceiling-Dual Gauge and Super-Block Renormalization to **every odd $q=2k+1\geq 3$** via the **Universal Core Gauge** $g_q(u):=(q^2-4)u+1$: 1. **Mod-8 Multiplier Lock:** $(q^2-4) \bmod 8=5$, so $\gcd(q^2-4, 2)=1$. 2. **Exact $s=2$ Step & Bridge Identity:** Whenever $4u_{r+1}=q^2u_r+1$, $4g_q(u_{r+1})=q^2g_q(u_r)$, and $(q^2-4)(q^2u_m+1)+4=q^2g_q(u_m)$. 3. **Universal Odd-Valuation Delayed-Action Death Trap ($v_2(g_q(u_0))=2m+1$):** If $g_q(u_0)=2\cdot 4^mV$ (V odd), then after m steps of $s=2$ ($g_q(u_m)=2q^{2m}V$), the next numerator satisfies $(q^2u_m+1) \bmod 4=2$ ($s_{r+m+1}=1$, guaranteed extinction). 4. **Universal Even-Valuation Exit-Rung Predictor ($v_2(g_q(u_0))=2m+2$):** If $g_q(u_0)=4^{m+1}V$ (V odd), then after m steps of $s=2$, writing $q^2u_m+1=4Y_m$ gives $(q^2-4)Y_m+1=q^{2m+2}V$, locking the exit rung s^* identically for **all odd $q\geq 3$**: - $V \bmod 4=3 \Longrightarrow (4Y_m) \bmod 16=8$ ($s^*=3$), - $V \bmod 8=5 \Longrightarrow (4Y_m) \bmod 32=16$ ($s^*=4$), - $V \bmod 8=1 \Longrightarrow (4Y_m) \bmod 32=0$ ($s^*\geq 5$).

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27. Eight Structural Angles (`SB.109–SB.120`): Adelic $\widehat{\mathbb{Z}}$ Decoupling, Real-First Parry Transfer, Flatto's $(3/2)^W$ Cylinder Bound, Base-6 Automaton, Two-Superblock Catalan–Pillai, Higher-Order Hankel Vanishing, Local Word-Powers V^m, & Rational Shadowing

In `sandbox_wave21_all_eight_angles_0_to_7.py`, we audited our proof architecture across eight distinct number-theoretic angles (`SB.109–SB.120`).

27.1 Angle 0 (`SB.109` , `adelic_profinite_odd_modulus_invertibility_and_archimedean_prime_floor`) — Odd-Modulus Solvability Across $\widehat{\mathbb{Z}}$ vs. $\mathbb{N}$ Divisor Floors

1. **Closed-Form Inverse Modulo Every Odd Modulus $M = 2m+1$:** Because $2(m+1) = (2m+1)+1 \equiv 1 \pmod{2m+1}$, $(2^T(m+1)^T) \bmod (2m+1) = 1$ for every $m \geq 1$ and $T \geq 0$.
2. **Universal Unique Solvability Across All Odd Moduli:** For **every** $m, T, P_0, A_K \in \mathbb{N}$, $P_{\text{sol}} = ((m+1)^T((3^T P_0 + A_K) \bmod M)) \bmod M$ is the **unique** residue in $[0, M)$ solving $(2^T P_{\text{sol}}) \bmod M = (3^T P_0 + A_K) \bmod M$. Analytically, this shows why finite-horizon congruence obstructions fail across $\widehat{\mathbb{Z}} \times \mathbb{R}$: odd prime moduli gain traction only when coupled to $\mathbb{N}$-specific archimedean order, such as $d \mid u_K \implies d \leq u_K \wedge d \cdot 2^{s_K} \leq P_K$.

27.2 Angle 1 (`SB.110` , `real_first_parry_transfer_map_invariance_and_dual_series_obstruction`) — The Real-First Parry Transfer Map on $[0, 1/2)$

Reversing direction from integer seeds to the connected interval $[0, 1/2)$ replaces the sparse survivor set with a forward-invariant interval map: 1. **Total Forward Invariance of $[0, 1/2)$:** Under the single-valued Parry map $F(r) = \frac{3}{2}r$ on $[0, 1/3)$ and $\frac{3}{2}r - \frac{1}{2}$ on $[1/3, 1/2)$, **every** rational point R_0/D with $1 \leq D$ and $2R_0 < D$ satisfies $2R_N < 2^N D$ for all $N \geq 0$. 2. **Dual Series Telescoping & Periodic Sign Obstruction:** Setting $\Psi_N = \sum_{k=0}^{N-1} 3^{N-1-k} 2^k e_k(r_0)$, we prove $R_N + D\Psi_N = 3^N R_0$ (rational companion side) and $2^N g_N = 3^N b_0 + \Psi_N$ (integer orbit side whenever $\forall k < N,\ \text{orbit}(b_0, k) \bmod 2 = e_k$), while $(3^L - 2^L)b_0 + \Psi_L = 0 \implies b_0 = 0 \wedge \Psi_L = 0$ ($L \geq 1$).

27.3 Angle 2 (`SB.111-SB.113`) — Formalization of Mahler–Fibonacci & Flatto's $(3/2)^W$ Parry Cylinder Survivor Bounds

Combining `SB.4` (`orbitParityRev_injOn_two_pow`) with pure-integer tree enumerations yields unconditional effective upper bounds on W-step surviving seeds in $[0, 2^W)$ (bridged directly to real Mahler Z-numbers over $\mathbb{R}$ in `SB.153` , Section 32.4): - **Theorem SB.111 (`unconditional_effective_fibonacci_survivor_power_saving_bound`):** `#survivors(< 2^W) ≤ mahlerFib (W + 2)` with logarithm-free power saving `mahlerFib (4k + 1) ≤ 8^k` and `mahlerFib (4k + 2) ≤ 2^(3k + 1)` ($O(X^{0.75}) = o(X)$). - **Theorem SB.112 (`parry_cylinder_tree_three_pow_sum_and_flp_effective_survivor_bound`):** Constructs the pure-integer Parry cylinder tree `parryCylTree W` ($B_w = 2^{W+1}|F^W(I(w))| \in \mathbb{N}$), proves **exact 3^W width conservation** (`parryWidthSum (parryCylTree W) = 3 ^ W`), and proves that for any set $\mathcal{S}_W \subseteq [0, 2^W)$ of seeds admitting rational W-step Parry companions $R_0/D \in [0, 1/2)$,

$$\boxed{2^W \cdot |\mathcal{S}_W| + 2 \cdot 2^W \ \leq\ 3^{W+1} \qquad \Big(|\mathcal{S}_W| \leq |\mathcal{C}_W| \leq 3 \cdot (3/2)^W - 2 = O\big(X^{\log_2(3/2)}\big) \approx O\big(X^{0.58496}\big)\Big),}$$

formalizing the combinatorial core of Flatto (1992) and Flatto–Lagarias–Pollington (1995) (extended to arbitrary finite sets of real Z-numbers in $[0, X)$ over $\mathbb{R}$ in `SB.153`). - **Theorem SB.113 (`post_horizon_top_bit_desert_and_two_word_difference_valuation`):** Proves `b_0 < 2^W ↔ b_0 / 2^W = 0` and proves that any two distinct W-bit survivors $b_1 < b_2 < 2^W$ must diverge strictly before step W ($n_0 = v_2(b_2 - b_1) < W$), never in a post-horizon tail $[W, W + M)$.

Bit-Horizon W	Exact W-Step Survivors $< 2^W$	Parry Cylinders $\lvert\mathcal{C}_W\rvert$	Fibonacci F_{W+2}	$2 \cdot (3/2)^W$	Max 2-Adic Fourier Ratio $\max_{1 \leq h \leq 15} \frac{\lvert S(h;W)\rvert}{\lvert\mathcal{C}_W\rvert}$
2	3	3	3	4.50	0.3333
4	8	8	8	10.12	0.3953
6	18	18	21	22.78	0.3821
8	40	40	55	51.26	0.2029
10	90	90	144	115.33	0.1093
12	201	201	377	259.49	0.1131
14	452	452	987	583.86	0.0411
16	1,018	1,018	2,584	1,313.68	0.0552

27.4 Angle 3 (`SB.114` , `base_six_terminating_digit_characterization_and_kempner_depth_law`) — Base-6 ($6 = 2 \times 3$) Joint Terminating Automaton

In Base 6, $\frac{3}{2} = 1.3_6$, $\frac{2}{3} = 0.4_6$, $\frac{1}{3} = 0.2_6$, and $\frac{1}{2} = 0.3_6$ all terminate in a single Base-6 digit: - Survival $r < 1/2$ is equivalent to first Base-6 fractional digit $f_1 \in \{0, 1, 2\}$; even steps satisfy $d_0 \in \{0, 2, 4\} \wedge f_1 \in \{0, 1\}$, while odd steps satisfy $d_0 \in \{1, 3, 5\} \wedge f_1 = 2$. - Every Kempner term terminates at exact Base-6 depth $T + 1$: $\frac{1}{3}(2/3)^T = \frac{2 \cdot 4^T}{6^{T+1}}$ (`3 * (2 * 4^T) * 3^T = 6^(T+1) * 2^T`).

27.5 Angle 4 (`SB.115-SB.116`) — Two-Superblock Catalan–Pillai Unit Equation, Unconditional Coprimality at $s^* \in \{3, 4\}$, & Conditional abc Cutoff

Chaining consecutive surviving Super-Blocks $W_{r+1} = 4^{m_1} V_1$ (exit $s^* \geq 3$) and $W_{r+1+m_1+1} = 4^{m_2} V_2$ (`4,165` $s^* = 3$ and `2,073` $s^* = 4$ transitions verified in Python, max abc-quality 1.3176): - **Theorem SB.115 (`two_superblock_catalan_pillai_transition_and_coprime_odd_cores`):**

$$\boxed{2^{2m_2 + s^*} V_2 + \left(3^{s^*-2} - 2^{s^*-2}\right) \ =\ 3^{2m_1 + s^*} V_1, \qquad \gcd(V_1, V_2) \ \Big|\ \left(3^{s^*-2} - 2^{s^*-2}\right),}$$

collapsing at $s^* = 3$ to the **Catalan–Pillai unit equation** $3^{2m_1+3}V_1 = 2^{2m_2+3}V_2 + 1$ with $\gcd(V_1, V_2) = 1$, and proving **unconditional coprimality** $\gcd(V_1, V_2) = 1$ **at** $s^* = 4$ **as well** (since $V_1 \bmod 5 \in \{1, 4\}$ by `SB.97` excludes $5 \mid V_1$). - **Theorem SB.116** (`superblock_abc_radical_compression_and_conditional_run_cutoff`): Proves every prime $p \mid (2^{e_2}V_2)D^*(3^{e_3}V_1)$ lies in $\{2, 3\} \cup \{p \mid D^*V_1V_2\}$, compressing 3^{2m_1} and 2^{2m_2} into $\{2, 3\}$ and bounding $3^{q(2m_1+s^*)}$ whenever an explicit power-radical inequality $(3^{e_3}V_1)^q \le K_{\text{abc}}(6D^*V_1V_2)^p$ is assumed.

27.6 Angle 5 (`SB.117`, `higher_order_hankel_row_reduction_and_superblock_vanishing_laws`) — Higher-Order $2 \times 2, 3 \times 3, 4 \times 4$ Hankel Row Reduction & Super-Block Vanishing

Along any sequence satisfying $2g_{k+1} = 3g_k + e_k$ (whose generating series $G(z) = \sum g_n z^n \in \mathbb{Z}[[z]]$ sits analytically on the Pólya–Carlson–Amice capacity boundary $R_\infty \cdot R_2 = 1$): - $2H_n^{(2)}$ and $4H_n^{(3)}$ reduce rows $1 \dots m-1$ to pure quadratic polynomials in $e_n, \dots, e_{n+2m-3}$, - $H_n^{(3)} = 0$ identically on every 4-zero run $(e_n, \dots, e_{n+3}) = (0, 0, 0, 0)$ ($s_r \ge 5$), and - $H_n^{(4)} = 0$ **identically on every 3-block** $s = 2$ **Super-Block** (`(1, 0, 1, 0, 1, 0)` or `(0, 1, 0, 1, 0, 1)`), for every starting seed.

27.7 Angle 6 (`SB.118`, `universal_local_word_power_gauge_divisibility_and_cutoff`) — Local Multi-Block Word-Power V^m Divisibility & Seed-Explicit Cutoff

For any multi-block word V of step sum $d \ge 1$ and numerator A_V ($5A_V \le 4(3^d - 2^d)$) repeating $m \ge 1$ times starting at block r (`9,114` local word powers verified in genuine orbits):

$$\boxed{2^{md} \,\Big|\, \big((3^d - 2^d)P_r + A_V\big) \qquad \text{and} \qquad 5 \cdot 2^{T_r + md} \;<\; 3^{T_r + d}(5P_0 + 4).}$$

Taking base-2 logarithms yields the seed-explicit horizon bound $(m - \log_2 3)\, d < (\log_2 3 - 1)\, T_r + \log_2((5P_0 + 4)/5)$. Once T_r is past the seed horizon ($5P_0 + 4 \le 2^{\varepsilon T_r}$), this elementary inequality rules out proportional cubes V^3 with $d \ge (0.42 + 0.71\varepsilon)T_r$ and squares V^2 with $d \ge (1.41 + 2.41\varepsilon)T_r$ (whereas in the pre-horizon regime $T_r \ll \log_2 P_0$, large seeds can realize local powers such as $V^3 = (3, 3)^3$ for $b_0 = 3{,}532{,}045$ at $T_1 = 3$).

27.8 Angle 7 (`SB.119`, `rational_remainder_recurrence_integer_escape_and_shadowing_ball_cutoff`) & 120-Theorem Synthesis (`SB.120`)

1. **Pure-Integer Exact Escape:** Every positive integer $p = 2^a(2t + 1)$ hits exact fractional part $\frac{1}{2}$ at step $a + 1$ (`(3^(a+1) * p) % 2^(a+1) = 2^a`).
2. **Rational Remainder Valuation Lock & Step Recurrence:** For $X_n = (3^n p) \bmod (2^n q)$ and $n > a = v_2(p)$, $2^a \mid X_n \wedge 2^{a+1} \nmid X_n$ ($X_n \ne 0$), and $X_{n+1} + 2^n q(g_n \bmod 2) = 3X_n$ across surviving steps.
3. **Quantitative Rational Shadowing Escape (`9,844` shadowing balls verified):** Whenever $X_n \in [2^{n-1}q + E,\ 2^n q - E]$ ($E \ge 1$), every rational candidate A/B satisfying $|qA - pB| \cdot 3^n < EB$ satisfies $2^{n-1}B < (3^n A) \bmod (2^n B)$ (forced out of $[0, 1/2)$ at step n).

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28. The 2-Adic Fourier Frontier (`SB.121-SB.124`): Conditional $O(X^{5/12})$ Sub-FLP Crossover vs. Barrier #6 (The Parseval Square-Root Entropy Canyon)

For each cylinder $(B, w) \in \text{parryCylTree}\ N$ with accumulator $\Psi_N(w)$, let $b_0(w) := (-\Psi_N(w) \cdot 3^{-N}) \bmod 2^N$ be its unique N-step seed residue, and define the analytic 2-adic character sum $\mathcal{S}(h; N) := \sum_{w \in \mathcal{C}_N} e^{2\pi i h b_0(w)/2^N} \in \mathbb{C}$. In `sandbox_wave23_fourier_frontier_and_parseval_barrier.py`, we evaluated exact radix-2 FFTs across all 2^N frequencies up to $N = 16$ ($65{,}536$ modes) and formalized `SB.121–SB.124`.

28.1 Theorem SB.121 (`two_sided_parry_entropy_sandwich_and_nonsplit_lower_bound`)

Proves that every node $(B, w) \in \text{parryCylTree}\ W$ satisfies $1 \le B \le 2^W$, sharpening `SB.112` into the **Two-Sided Pure-Integer Parry Entropy Sandwich**:

$$\boxed{3^W \;\le\; 2^W \cdot |\mathcal{C}_W| \;\le\; 3^{W+1} - 2 \cdot 2^W.}$$

Furthermore, because every right child produced by a split at step k satisfies $3(3B - 2^{k+1}) \le 2^{k+2}$, **no right child can split at step** $k + 1$, forcing the non-splitting cylinder count $U_k = 2|\mathcal{C}_k| - |\mathcal{C}_{k+1}|$ to satisfy $|\mathcal{C}_k| \le U_k + U_{k+1}$ and $3^k \le 2^k(U_k + U_{k+1})$.

28.2 Theorem SB.122 (`splitting_pair_half_period_antiphase_and_odd_parseval_identity`)

Establishes the modular antiphase and real-space L^2 identities underlying odd-frequency cancellation: 1. **Half-Period Antiphase Shift:** Whenever a cylinder at depth $N - 1$ splits into two children ($e_{N-1} \in \{0, 1\}$), their unique residues $b_{\text{left}}, b_{\text{right}} \in [0, 2^N)$ satisfy $b_{\text{right}} = (b_{\text{left}} + 2^{N-1}) \bmod 2^N$ and $((2m + 1)b_{\text{right}}) \bmod 2^N = ((2m + 1)b_{\text{left}} + 2^{N-1}) \bmod 2^N$ (shifting every odd character by π, so $e^{\pi i} + 1 = 0$ analytically). 2. **Real-Space Half-Period Difference Identity:** For any $\{0, 1\}$-indicator f on $[0, 2^N)$ with $H = 2^{N-1}$, $\sum_{y<H}(f(y) - f(y + H))^2 + 2\sum_{y<H} f(y)f(y + H) = \sum_{y<H}(f(y) + f(y + H))$, which by Plancherel's theorem on $\mathbb{Z}/2^N\mathbb{Z}$ (informal) equates $\frac{1}{2^{N-1}} \sum_{h \text{ odd}} |\mathcal{S}(h; N)|^2 = 2|\mathcal{C}_{N-1}| - |\mathcal{C}_N| = U_{N-1}$.

Depth N	$\lvert\mathcal{C}_N\rvert$	Non-Splits U_{N-1}	$\max_{h \text{ odd}} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert}$	$\text{RMS}_{h \text{ odd}} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert} = \frac{\sqrt{U_{N-1}}}{\lvert\mathcal{C}_N\rvert}$	$\max_{h>0} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert}$	$\arg\max_{h>0}$
4	8	2	0.24520	0.17678	0.39528	4

Depth N	$\lvert\mathcal{C}_N\rvert$	Non-Splits U_{N-1}	$\max_{h\text{ odd}} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert}$	$\mathrm{RMS}_{h\text{ odd}} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert} = \frac{\sqrt{U_{N-1}}}{\lvert\mathcal{C}_N\rvert}$	$\max_{h>0} \frac{\lvert\mathcal{S}(h;N)\rvert}{\lvert\mathcal{C}_N\rvert}$	$\arg\max_{h>0}$
6	18	6	0.28543	0.13608	0.40062	16
8	40	14	0.21149	0.09354	0.42180	224
10	90	30	0.21672	0.06086	0.41211	128
12	201	67	0.20064	0.04072	0.40706	512
14	452	152	0.18468	0.02728	0.40967	14,336
16	1,018	338	0.14184	0.01806	0.40681	$57{,}344 = 7 \cdot 2^{13}$

(Note: High-2-power modes with $2^{N-W} \mid h$ satisfy $(h \cdot 2^W) \bmod 2^N = 0$ by `SB.151`, so $\widehat{\mathbf{1}_{[0,2^W)}}(-h) = 0$ identically.)

28.3 Theorem SB.123 (`conditional_fourier_sub_flp_exponent_and_seventeen_twelfths_crossover`) — Conditional Reduction to $O(X^{5/12}) \approx O(X^{0.4167})$

In the intermediate post-bit-length window $W < N < 2W$, balancing the main term $2^W(3/4)^N$ against square-root Fourier error $(3/2)^{N/2}$ crosses at $N^* = \left(1 + \frac{\log(3/2)}{\log(8/3)}\right)W \approx 1.4134\,W$. Using the rational convergent $17/12 \approx 1.4167$ ($3^{17} = 129{,}140{,}163 < 134{,}217{,}728 = 2^{27}$), `SB.123` proves that for every $k \in \mathbb{N}$ and any count S_{card} satisfying **both** (i) the Beurling–Selberg majorant inequality $2^{5k} S_{\text{card}} \le 2|\mathcal{C}_{17k}| + 2^{5k} E$ and (ii) the square-root error bound $2^{17k} E^2 \le C_{\text{err}}^2 \cdot 3^{17k}$:

$$\boxed{S_{\text{card}} \;\le\; (6 + C_{\text{err}}) \cdot 2^{5k} \;=\; O\Big(X^{5/12}\Big) \;\approx\; O\big(X^{0.4167}\big) \;\ll\; O\big(X^{0.58496}\big)\,.}$$

28.4 Theorem SB.124 (`barrier_six_parseval_sqrt_canyon_and_124_theorem_synthesis`) — Barrier #6: The Parseval Square-Root Floor

Why can't an unsigned (L^1 triangle-inequality) Fourier bound drive the survivor count below 1 at $N = 3W$? 1. `SB.124` proves in $\mathbb{N}$ that for every k, either $3^k \le 2^{k+1} U_k$ or $3^k \le 2^{k+1} U_{k+1}$, and for every $W \ge 2$, $3^{3W} < 2^{5W}$ while $2^{4W-3} < 3^{3W-2}$. 2. Analytically (via Plancherel), this forces $\max_{h\text{ odd}} |\mathcal{S}(h;k')|^2 \ge U_{k'-1} \ge \frac{3^k}{2^{k+1}} \to \infty$ for $k' \in \{k+1, k+2\}$ (and by unitary invariance, the same L^2 norm floor holds in every orthonormal basis of $\mathbb{C}[\mathbb{Z}/2^N\mathbb{Z}]$, including Walsh–Hadamard). At $N = 3W$, the main term $|\mathcal{C}_{3W}| \cdot 2^{-2W} \le 3(27/32)^W$ falls strictly below 1 for all $W \ge 7$ (at $W = 2, N = 6$ it is $18/16 = 9/8$), whereas the unsigned L^1 triangle-inequality error grows exponentially unless signed phase cancellation is retained, as illustrated at $W = 5$ ($2^W = 32$):

Horizon N	Actual Survivors < 32	Main Term $\lvert\mathcal{C}_N\rvert \cdot 2^{5-N}$	L^1 Triangle-Inequality Error	Signed Fourier Error
5	12	12.0000	0.00	$+0.0000$
7	10	6.7500	6.59	$+3.2500$
9	6	3.7500	12.22	$+2.2500$
11	2	2.0938	19.29	-0.0938
13	1 ($b_0 = 0$)	1.1797	28.87	-0.1797
15	1 ($b_0 = 0$)	0.6621	41.48	$+0.3379$

29. Post-Parseval Frontiers (`SB.125–SB.132`): Multiplicative $\langle 3\rangle$-Coset Flattening, 3-Point Window Scaling, Skew-Product Anti-Recurrence, & Conditional Quadratic abc

In `sandbox_wave24_paths_a_and_b_deep_exploration.py`, we formalized `SB.125–SB.132` across both post-Parseval directions.

29.1 Path A (`SB.125–SB.127`) — Multiplicative $\langle 3\rangle$-Subgroup Order, Concatenation Cocycle, & Coset Half-Sum Identity

1. **Exact LTE_2 Order 2^{K-2} & Disjoint $\pm\langle 3\rangle$ Partition (`SB.125`, `lte2_order_of_three_and_pm_cyclic_partition_of_odd_residues`):** For every $K = m + 4 \ge 4$, $3^{2^{m+1}} \bmod 2^{m+4} = 2^{m+3} + 1 \ne 1$ and $3^{2^{m+2}} \bmod 2^{m+4} = 1$, while $+\langle 3\rangle \bmod 8 \subseteq \{1, 3\}$ and $-\langle 3\rangle \bmod 8 \subseteq \{5, 7\}$ are strictly disjoint, partitioning all 2^{K-1} odd residues in $(\mathbb{Z}/2^K\mathbb{Z})^\times$ into two cosets.
2. **Two-Stage Parry Concatenation Cocycle & $\times 3^{N_1}$ Frequency Rotation (`SB.126`, `two_stage_parry_concatenation_cocycle_and_frequency_rotation`):** Across $N_1 + N_2$ steps, $\Psi_{N_1+N_2} = 3^{N_2}\Psi_{N_1} + 2^{N_1}\Psi_{N_2}$, and $(3^{N_1} b_{\text{total}}) \bmod 2^{N_1+N_2} = (3^{N_1} b_{\text{pref}} + 2^{N_1} b_{\text{cont}}) \bmod 2^{N_1+N_2}$, rotating the continuation residue along $\langle 3\rangle \le (\mathbb{Z}/2^{N_2}\mathbb{Z})^\times$.
3. **Conjugate-Symmetric Coset Split (`SB.127`, `conjugate_symmetric_coset_split_and_exact_orbit_mean_identity`):** Proves in $\mathbb{N}$ that for any disjoint decomposition $C_{\text{pos}} \sqcup C_{\text{neg}}$ mapped bijectively by $x \mapsto 2^K - x$ and any symmetric weight $M_{\text{sq}}(2^K - x) = M_{\text{sq}}(x)$, $(C_{\text{pos}} \cup C_{\text{neg}}).\text{sum}\ M_{\text{sq}} = 2 \cdot C_{\text{pos}}.\text{sum}\ M_{\text{sq}}$. Analytically, since $|\mathcal{S}(2^K - h; K)|^2 = |\mathcal{S}(h; K)|^2$, this locks the $\langle 3\rangle$-orbit mean-square of every odd frequency h_{odd} to U_{K-1} (with empirical 4th-moment ratio $\mathbb{E}[|\mathcal{S}|^4]/U^2 \le 5.97$ at $K = 15$).

29.2 Path B1 (`SB.128–SB.129`) — 2-Point Divergence $(9/8)^N$ vs. 3-Point Scaling $(27/32)^N$ & Zero-Triple Uniqueness

- **Theorem SB.128 (`three_point_window_scaling_decay_vs_two_point_divergence`):** Proves the cardinality bounds $8^N < 9^N \le 4^N|\mathcal{C}_N|^2$ ($|\mathcal{C}_N|^2/2^N \ge (9/8)^N \to \infty$) vs. $8^N|\mathcal{C}_N|^3 < 27 \cdot 27^N$, and proves $27 \cdot 4^W \cdot 27^{9W} < 32^{9W}$ for all $W \ge 24$ (showing that under a uniform-density heuristic $2^{2W}|\mathcal{C}_N|^3/2^{2N}$, the heuristic 3-point window count drops below 1 at $N = 9W$; proving non-clustering of actual triples remains open).
- **Theorem SB.129 (`zero_three_point_window_triples_implies_at_most_one_positive_survivor`):** Proves unconditionally that whenever a finite set $S \ni 0$ has no ordered triple $x < y < z$, $|S| \le 2$ and $|S \setminus \{0\}| \le 1$ (at most one positive element). The $W = 3$ ($2^W = 8$) empirical census across all $|\mathcal{C}_N|$ cylinders is shown below alongside the uniform-density heuristic $64(27/32)^N$:

Depth N	Survivors in $[0, 8)$	All Pairs $0 < y - x < 8$ on $[0, 2^N)$	Ordered Triples in $[0, 8)$	All Triples $0 \le x < y < z, z - x < 8$ on $[0, 2^N)$	Uniform Heuristic $64(27/32)^N$
3	5	10	10	10	38.44
7	3	35	1	17	19.48
11	2	47	0	7	9.87
15	1	70	0	1	5.00
16	1 ($b_0 = 0$)	72	0	0	4.22

29.3 Path B2 (`SB.130`, `skew_product_strict_growth_and_poincare_anti_recurrence_law`) — Deterministic Skew-Product Growth & Quantitative Gap Bound

For a measure-preserving system on a finite-measure space, Poincaré recurrence guarantees almost-everywhere returns; by contrast, on $\mathbb{N}_{>0}$ the Mahler ceiling step is strictly expanding. `SB.130` proves $b < \mathrm{step}(b)$ for all $b \ge 1$, $\mathrm{orbit}(b_0, n_1) < \mathrm{orbit}(b_0, n_2)$ for all $n_1 < n_2$, and whenever $n_1 < n_2$ share L future parity bits (`12,935` pairs verified in Python), `2^L ≤ orbit b_0 n_2 - orbit b_0 n_1 < orbit b_0 n_2` (bounding shared-word length $L < \log_2 \mathrm{orbit}(b_0, n_2)$ rather than forbidding short pre-horizon repeats).

29.4 Path B3 (`SB.131–SB.132`, `quadratic_abc_two_superblock_mutual_run_cutoff` & `grand_paths_a_and_b_132_theorem_semigroup_breakout_synthesis`)

Assuming an explicit quadratic abc-type inequality $3^{2m_1+3}V_1 \le K_2(6V_1V_2)^2$ on $2^{2m_2+3}V_2 + 1 = 3^{2m_1+3}V_1$ (`SB.115`), `SB.131` eliminates V_2 to prove $2^{4m_2+6} < 36K_2V_1^3 \cdot 3^{2m_1+3}$ and derives simultaneous bounds $3^{2m_1+3} \le 36K_2V_{\max}^3$ and $2^{2m_2+3} < 36K_2V_{\max}^4$ whenever $V_1, V_2 \le V_{\max}$.

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30. Standalone Method 1–3 Obstructions, Prime Supply, & The $159/160$ Mod-320 Even-Gauge Pinch (`SB.133–SB.140`)

In `sandbox_wave25_method2_ideas_and_real_gauge_pinch.py`, we formalized the quantitative limits of three standalone strategies (`SB.133–SB.135`) and four fusion mechanisms (`SB.136–SB.140`).

30.1 Standalone Method 1–3 Quantitative Obstructions (`SB.133–SB.135`)

1. **Method 1 Pigeonhole Cylinder Excess (`SB.133`, `method_one_pigeonhole_cylinder_excess_obstruction`):** At $M = 12k$ steps ($12k + 1$ orbit points), `SB.130` bounds shared parity prefixes by $L = 7k$ bits (`2^{7k} = 128^k`). Because $17 \cdot 128 = 2176 < 2187 = 3^7$, $|\mathcal{C}_{7k}| \ge 17^k > 12k + 1$ for all $k \ge 1$, leaving exponentially many distinct cylinders available without collision.
2. **Method 2 Product Excess vs. Classical abc (`SB.134`, `method_two_abc_generic_radical_excess_identity_and_bound`):** In $2^{S_0}u_K = 3^{T_K}u_0 + R_0$ (`SB.72`, with $s_0 \ge 2$ and $\forall k < M,\ s_{1+k} \ge 2$), $R_0 = 3^{d_0} + \mathrm{renewalNumerator} \ge 3^{d_0}$ ($d_0 = T_K - s_0$), forcing the **product** inequalities $3^{T_K+d_0}u_0 < 2^{S_0}(u_KR_0)$ and `2^{S_0} u_K < u_K R_0` whenever $2^{S_0} < 3^{d_0}$. When u_K and R_0 are both squarefree and coprime ($\gcd(u_K, R_0) = 1$), $\mathrm{rad}(ABC) \ge u_KR_0 \asymp C^{1.3691} > C$ exceeds C; note, however, that $\gcd(u_K, R_0)$ divides u_0 and can be non-trivial when $u_0 > 1$ (for example, $b_0 = 2469$ yields $u_3 = 173 \cdot 463$ and $R_0 = 5 \cdot 463$ sharing the factor $463 \mid u_0$), so passing to the primitive abc triple requires dividing out $\gcd(u_K, R_0)$.
3. **Method 3 Shift-Average Conservation vs. Occupied Shifts (`SB.135`, `method_three_markov_shift_average_vs_bad_shift_occupancy`):** Proves $\sum_{a<M}\sum_{j<L} f((a+j) \bmod M) = L\sum_{x<M} f(x)$, while every occupied residue $b \in \mathcal{C}_{3W}$ yields $\sum_{j<L} f((b+j) \bmod M) \ge 1$.

30.2 Prime Supply ≥ 5, Short-Recovery Ban (`1 ≤ c ≤ 5`), & Mod-320 Even-Gauge Pinch (`SB.136–SB.140`)

1. **Interior Odd-Core Prime Supply ≥ 5 (`SB.136`, `interior_odd_core_coprime_to_six_and_prime_divisor_ge_five`):** For every interior block $r + 1 \ge 1$ (`1 ≤ s r`), $\gcd(u_{r+1}, 6) = 1$, every prime $p \mid u_{r+1}$ (`Nat.Prime p`) satisfies $p \ge 5$, and $u_{r+1} > 1 \implies u_{r+1} \ge 5$.
2. **Dual Multiplicative-to-Additive Excess Sandwich (`SB.137`, `dual_multiplicative_euler_product_to_additive_kempner_sandwich`):** Along any chain with $s_0 \ge 2$ and $\forall k < M,\ s_{1+k} \ge 2$, sandwiches $\prod_{k=0}^{K-1}(3^{s_k}u_k + 1) - 3^{T_K}\prod_{k=0}^{K-1}u_k = R_0\prod_{k=1}^{K-1}u_k$ between $3^{d_0}\prod_{k=1}^{K-1}u_k$ and $\frac{1}{5}\cdot 3^{T_K}\prod_{k=1}^{K-1}u_k$.
3. **Rational $[0, 1/2)$ Escape of All Five Short Post-`2` Recoveries `c ∈ {1, 2, 3, 4, 5}` (`SB.138`, `real_half_interval_short_recovery_ban_two_two_through_two_five`):** Proves $\Psi_{3+m}(1, 0, 1, 0^m) = 13 \cdot 3^m$, forcing $(13 \cdot 3^c + 4 \cdot 2^c)D \le 27 \cdot 3^cR_0$ whenever a rational companion $2R_0 < D$ (`1 ≤ D`) has bit 1 at step $2 + c$. For **all five** values $c \in \{1, 2, 3, 4, 5\}$, the threshold exceeds $\frac{1}{2}$ ($\frac{47}{81}, \frac{133}{243}, \frac{383}{729}, \frac{1117}{2187}, \frac{3287}{6561} > \frac{1}{2}$), whereas $c = 6$ ($\frac{9733}{19683} < \frac{1}{2}$) is the sharp minimum recovery rung.
4. **Mod-320 Even-Gauge Pinch (`SB.139–SB.140`, `real_coupled_gauge_mod64_and_mod320_unique_even_survivor_class`):** Along any `ValidBlockChain b_0 s u (r + 3)` with $2 \le s_r, s_{r+1} = 2$, and $6 \le s_{r+2}$ (a premise discharged directly from any surviving real companion over $\mathbb{R}$ in `SB.154`, Section 32.4, and

empirically verified across `200,000` odd seeds and `267` surviving interior even-W blocks with `0` exceptions):

$$\boxed{W_{r+1} \bmod 64 = 4, \qquad W_{r+1} \bmod 320 = 4, \qquad W_{r+1} = 4V \text{ with } V \bmod 80 = 1.}$$

Among all 160 even residue classes modulo 320, 159 (99.375%; $31/32$ incremental beyond the mod-5 lock $W_{r+1} \equiv 4 \pmod{10}$) are excluded, leaving only $W_{r+1} \equiv 4 \pmod{320}$.

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31. Parry Kneading Renewal, $O(N^2)$ Fourier DP Out to $N = 300$, 3-Adic Phase Debt, & Origin Laplace–Poisson Threshold Criteria (`SB.141–SB.148`)

In `sandbox_wave26_twisted_operator_and_laplace_partition.py`, we formalized `SB.141–SB.148` and implemented an exact $O(N^2)$ dynamic-programming evaluator for individual 2-adic character modes $\mathcal{S}(h; N)$.

31.1 Exact Parry Kneading Renewal, Reproducible $N = 300$ DP Benchmark, & 3-Adic Phase Debt (`SB.141–SB.144`)

1. **Exact Parry Kneading Spine Cylinder-Count Renewal (`SB.141` , `parry_kneading_spine_renewal_decomposition_of_cylinder_counts`):** Whenever the rightmost (`1` -branch) kneading spine $B_j = \mathrm{kneadingSpineB}\ j$ splits at depth j (`2^{j+1} < 3 B_j` , first occurring at $m = j + 1 \in \{1, 3, 9, 12, 15, \dots\}$), its `0` -child has scaled width 2^{j+1} (a full renewal copy of $[0, 1/2)$). `SB.141` proves in $\mathbb{N}$:

$$\boxed{|\mathcal{C}_N| \;=\; 1 \;+\; \sum_{\substack{0 \le j < N \\ 2^{j+1} < 3B_j}} |\mathcal{C}_{N-1-j}|.}$$

   Analytically (informal), this induces the character-sum renewal $\mathcal{S}(h; N) = e^{2\pi i h b^{(N)}_{\mathrm{spine}}/2^N} + \sum_{m_r \le N} e^{2\pi i h b^{(r)}_{\mathrm{pref}}/2^N} \mathcal{S}((3^{-m_r} h) \bmod 2^{N-m_r};\ N - m_r)$, evaluating any single odd mode h in $O(N^2)$ operations along its 3^{-1}-orbit (verified to 10^{-7} precision against a full 4096-point FFT across all 2048 odd modes at $N = 12$ in `test_move1_quadratic_dp_and_hensel_300`). Across structured modes $h = 1$, $h = 3^{\lfloor N/2 \rfloor}$, and a 16-beam upward Hensel search out to $N = 300$, we observe finite-N contraction $\rho = |\mathcal{S}(h; N)|^{1/N} / |\mathcal{C}_N|^{1/N} < 1$:

Horizon N	$\rho(h = 1)$	$\rho(h = 3^{\lfloor N/2 \rfloor})$	ρ(16-Beam Hensel)	$\Lambda_{\mathrm{Hensel}} = \lvert\mathcal{S}\rvert^{1/N}$
50	0.73857	0.88001	0.93624	1.41675
100	0.81069	0.85968	0.94615	1.42546
150	0.81041	0.88881	0.94933	1.42817
200	0.77883	0.90130	0.95289	1.43248
250	0.79721	0.87320	0.95174	1.43013
300	0.78601	0.90043	0.95111	1.42876

1. **Phase Syzygy & Multi-Scale 3-Adic Debt Law (`SB.142–SB.143`):**
2. `SB.142` : Proves $(3b_{\mathrm{pref}}(3) + 1) \bmod 2^N = 0$, $(27b_{\mathrm{pref}}(9) + 13) \bmod 2^N = 0$, and `(9 * b_pref3) % 2^N = (13 * b_pref2) % 2^N (N ≥ 3)` .
3. `SB.143` : Proves that for $1 \le s < K$, $1 \le y_K < 2^K$, and $1 \le y_{\mathrm{sub}} < 2^{K-s}$ satisfying $(3^s y_{\mathrm{sub}}) \bmod 2^{K-s} = y_K \bmod 2^{K-s}$, whenever $d_K + 3^s d_{\mathrm{sub}} < 2^{K-s}$ we have exact integer equality `d_K = 3^s * d_sub` , forcing `3 ^ (blockSteps step_s L) ≤ d_0 < 2^K` across any L-step small-angle chain ($\Delta K < (\log_3 2) K < 0.63093\, K$).
4. **Conditional $8:7$ Spectral-Gap Reduction (`SB.144` , `pointwise_spectral_gap_to_unconditional_sub_flp_survivor_exponent`):** At $W = 7k$, $N = 8k$ ($k \ge 1$), since $3^8 < 13 \cdot 2^9$ and $7^8 < 15 \cdot 5^8$, any count S_{card} satisfying **both** (i) $2^k S_{\mathrm{card}} \le |\mathcal{C}_{8k}| + 2^k E$ and (ii) $5^{8k} E \le C_0 \cdot 7^{8k}$ ($\Lambda \le 7/5 = 1.40$) obeys `S_card ≤ 3 * 13^k + C_0 * 15^k` ($O(X^{\log_2(15)/7}) \approx O(X^{0.55813})$, strictly below `SB.133` 's $17^k \le |\mathcal{C}_{7k}|$). Note that the empirical $N = 300$ beam rate $\Lambda_{\mathrm{Hensel}} \approx 1.4288$ exceeds $7/5 = 1.40$ (at $\Lambda \approx 1.429$, a closer horizon split $N \approx 1.07W$ rather than $8:7$ is needed, and a uniform all-frequency proof remains open).

31.2 Pure-$\mathbb{N}$ Euler Sandwich, One-Sided $4/3$ Origin Partition Criteria, & Discrete Poisson Core Positivity (`SB.145–SB.148`)

1. **Pure-$\mathbb{N}$ Bernoulli & Two-Sided Euler Sandwich (`SB.145` , `pure_nat_euler_sandwich_for_origin_laplace_kernel`):** Proves in $\mathbb{N}$ (via strict decrease of $((n+1)/n)^{n+1}$) that for every $m \ge 2$:

$$\boxed{2(m-1)^m \;<\; m^m \quad \left(\left(\frac{m-1}{m}\right)^m < \frac{1}{2}\right) \qquad \text{and} \qquad m^{m-1} \;<\; 3(m-1)^{m-1} \quad \left(\left(\frac{m-1}{m}\right)^{m-1} > \frac{1}{3}\right).}$$

2. **Two One-Sided $4/3$ Origin Partition Threshold Criteria (`SB.146` , `origin_laplace_partition_two_way_mahler_threshold_equivalence`):** For rational parameters $L \ge 2$ (e.g., $L = 2^W$), $M_{\max} \in \mathbb{N}$, and $0 \in S$ with $\mathrm{laplacePartitionNum}(L, M_{\max}, S) = \sum_{x \in S} (L-1)^x L^{M_{\max} - x}$:
3. **Positive Witness Below L Forces $> 4/3$:** If $\exists b \in S$ with $1 \le b < L$ and $b \le M_{\max}$, then `4 * L^M_max < 3 * laplacePartitionNum L M_max S` ($\mathcal{Z}^{\mathrm{rat}}_N(L) > 4/3$).
4. **Empty Prefix $[1, kL)$ Forces $< 4/3$:** If $1 \le k$, $\forall b \in S \setminus \{0\}$, $kL \le b \le M_{\max}$, and $3|S \setminus \{0\}| \le 2^k$, then `3 * laplacePartitionNum L M_max S < 4 * L^M_max` ($\mathcal{Z}^{\mathrm{rat}}_N(L) < 4/3$). *(Intermediate configurations with survivors in $[L, kL)$ are not covered by either one-sided criterion.)*

5. **Discrete Poisson Denominator-Core Positivity (`SB.147–SB.148`, `discrete_poisson_kernel_strict_real_part_positivity`):** Proves over $\mathbb{Z}$ that for all $0 \le r < D$ and $-D \le C \le D$ ($C/D = \cos\theta \in [-1,1]$), $0 < D^2 - 2rC + r^2 < 2(D^2 - rC)$, so the normalized Poisson core satisfies $\mathrm{Re}\left(\frac{1}{1-(r/D)e^{-i\theta}}\right) > \frac{1}{2}$ (and multiplying by $1-(r/D)^M > 0$ gives $\mathrm{Re}\,\widehat{K}(-h) > \frac{1-(r/D)^M}{2} > 0$). At $W = 4$ ($L = 2^4 = 16$, ratio $15/16$), $\mathcal{Z}_N^{\mathrm{rat}}(16)$ drops from `2.9396` ($N = 10$) to `1.0391 < 4/3` at $N = 12$ once $b_0 = 4$ dies, while the unsigned L^1 Poisson error continues growing (`10.42 -> 15.79 -> 33.48`).

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32. Exact $(N+1)$ Width Compression, Barrier #7 Rogue-Reset Uniqueness, & Full Real-Number ($\mathbb{R}$) Z-Number Bridges (`SB.149–SB.154`)

In `sandbox_wave27_capstone_rank_compression_and_barrier7.py` and the closing section of `SemigroupBreakoutTheorems.lean`, we formalized `SB.149–SB.154`, including direct real-number ($\mathbb{R}$) bridges for both Flatto's $(3/2)^W$ Z-number bound and the mod-320 even-gauge pinch.

32.1 Theorem SB.149 (`parry_cylinder_tree_exact_n_plus_one_width_class_compression`) — Exact $(N+1)$ Scaled-Width Bijection

Although `(parryCylTree N).length` grows as $\Theta((3/2)^N)$, every cylinder was born at its most recent full $[0,1/2)$ renewal at some depth $N - r$ ($0 \le r \le N$) and then evolved along the kneading spine for r steps: 1. **Universal Oddness of `kneadingSpineB`:** Every spine width $B_r = \mathrm{kneadingSpineB}\ r$ satisfies `1 ≤ B_r ∧ B_r % 2 = 1`, so the shifted widths $2^{N-r}B_r$ ($0 \le r \le N$) have pairwise distinct 2-adic valuations $N, N-1, \ldots, 0$. 2. **Exact Finset Bijection & Linear Cardinality (`∀ N`):**

$$\boxed{\left\{ B \;:\; (B,w) \in \mathrm{parryCylTree}\ N \right\} \;=\; \left\{ 2^{N-r}B_r \;:\; 0 \le r \le N \right\}, \qquad \mathrm{card} \;=\; N+1.}$$

(Informally, because continuation subtrees depend only on starting scaled width, this bounds the rank of the two-stage continuation operator across `SB.126` by $N_1 + 1$.)

32.2 Theorem SB.150 (`upward_hensel_phase_step_dichotomy_and_barrier7_rogue_reset_law`) — Upward Hensel Phase Dichotomy & Barrier #7

1. **Upward Hensel Step Dichotomy:** For all $k \ge 1$, $1 \le y_k < 2^k$, and $1 \le y_{\mathrm{next}} < 2^{k+1}$ satisfying $y_{\mathrm{next}} \bmod 2^k = (3y_k) \bmod 2^k$, setting $d_k = \min(y_k, 2^k - y_k)$ and $d_{\mathrm{next}} = \min(y_{\mathrm{next}}, 2^{k+1} - y_{\mathrm{next}})$, **either** `8 * d_k ≥ 2^k ∨ 8 * d_next ≥ 2^(k+1)` (at least one endpoint is a $\ge 1/8$ phase step), **or** `d_next = 3 * d_k`.
2. **Barrier #7 (Multiplicative-to-Archimedean Frequency Uncertainty & Per-Quotient Uniqueness):** Bridging $\langle 3 \rangle$-coset cancellation (`SB.127`) to short archimedean bands $|h| \le 2^{N-W}$ or arbitrary single modes requires ruling out two-variable rogue resets $|3^A d - q \cdot 2^K| < R$ with variable $d < 2^{0.40K}$ (where Matveev loses by $\log K$ due to the linear Weil height of d). Elementary subtraction (`SB.150.2`) proves over $\mathbb{Z}$ that whenever $2R \le 3^A$, **at most one integer d can exist per quotient q** (`d₁ = d₂`).

32.3 Theorem SB.151 (`dyadic_frequency_factorization_and_box_high_power_vanishing`) & Theorem SB.152 (`grand_152_theorem_mahler_semigroup_breakout_capstone_synthesis`)

For $W \le N$ and $1 \le h < 2^N$: if $2^{N-W} \mid h$, then $(h \cdot 2^W) \bmod 2^N = 0$ and $h \bmod 2^N \ne 0$ (full-period box annihilation); if $\neg(2^{N-W} \mid h)$, then $h = 2^v h_{\mathrm{odd}}$ with $v < N - W$, $h_{\mathrm{odd}} \bmod 2 = 1$, and $h_{\mathrm{odd}} < 2^{N-v}$, synthesized with `SB.112, SB.121, SB.139, SB.141, SB.149, SB.150` in `SB.152`.

32.4 Theorems SB.153 & SB.154 — Machine-Checked Bridges Directly Over $\mathbb{R}$

To close the gap between natural-number Parry cylinder theorems and real Mahler Z-numbers without informal steps, we formalized two final theorems over $\mathbb{R}$ using `ValidCompanionUpTo`, `ValidCompanionForever`, and `Int.fract`: 1. **Theorem SB.153 (`real_z_number_uniqueness_parry_cylinder_bridge_and_flatto_bound`):** - *Mahler's Uniqueness Lemma over $\mathbb{R}$:* `∀ (b_0 : ℕ) (α β : ℝ), ValidCompanionForever b_0 α → ValidCompanionForever b_0 β → α = β` (at most one real Z-number per unit interval $[b_0, b_0+1)$). - *Direct Real `parryCylTree` Coverage:* `ValidCompanionUpTo b_0 α W → ∃ B, (B, orbitParityRev b_0 W) ∈ parryCylTree W ∧ (2 : ℝ)^(W+1) * fracPart b_0 α W < (B : ℝ)`. - *Unconditional Real Z-Number Bound on $[0, X)$:* For **every** $W \in \mathbb{N}$, $X \le 2^W$, and finite set `Z_set : Finset ℝ` satisfying `∀ α ∈ Z_set, 0 ≤ α ∧ α < X ∧ ∀ n : ℕ, Int.fract (α * (3 / 2 : ℝ) ^ n) < 1 / 2`,

$$\boxed{\mathrm{Z_set.card} \;\le\; (\mathrm{parryCylTree}\ W).\mathrm{length} \qquad \text{and} \qquad 2^W \cdot \mathrm{Z_set.card} + 2 \cdot 2^W \;\le\; 3^{W+1}.}$$

2. **Theorem SB.154 (`unconditional_real_companion_mod320_even_gauge_pinch`):** Along any `ValidBlockChain b_0 s u (r + 3)` admitting a real companion `ValidCompanionUpTo b_0 α (blockSteps s (r + 3) + 1)` (hence for every `ValidCompanionForever b_0 α`) with `2 ≤ s r` and `s (r + 1) = 2`, tracking `fracPart` across blocks $r+1$ and $r+2$ proves $8 \cdot 2^{s(r+2)} < 3^{s(r+2)}$, discharging `6 ≤ s (r + 2)` automatically over $\mathbb{R}$ and proving `W_{r+1} % 64 = 4 ∧ W_{r+1} % 320 = 4 ∧ V % 80 = 1` unconditionally.

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33. Architectural Conclusion: What Is Machine-Checked and Why Seven Structural Barriers Guard $Z = \emptyset$

Across our 226-file root library (`MAHLER_FORMALIZATION_PAPER.pdf`) and this 154-theorem breakout library (`SemigroupBreakoutTheorems.lean`), **Mahler's full $3/2$ conjecture ($Z = \emptyset$) remains open**. The table below separates what is formally verified in Lean 4 from the analytical and external Diophantine

barriers that explain why each standalone method stalls short of $Z = \emptyset$:

#	Structural Barrier	Machine-Checked in Lean 4 (`SemigroupBreakoutTheorems.lean` & Root)	Analytic / External Literature Mechanism
1	**Adelic $\widehat{\mathbb{Z}} \times \mathbb{R}$ Decoupling**	`SB.104` (2^{S_K+1}-periodicity vs. $P_0 < 2^{S_K}$ uniqueness), `SB.109` (closed-form inverse & unique root mod every $2m+1$; $\mathbb{N}$ divisor floor).	Meta-Theorem 2.1: finite-horizon congruences and companion intervals decouple P_0 from h_0 across $\widehat{\mathbb{Z}} \times \mathbb{R}$.
2	**Baker–Matveev Linear Height Loss**	`SB.67`, `SB.71` (3^{s_r}-upgraded linear form pinch & repeated-core $u_{r+M} = u_r$ bound; `BakerMatveevFramework.lean`).	Matveev (2000): moving odd cores $u_K \sim (3/2)^{T_K}$ have linear Weil height, losing a $\log T_K$ factor unless cores repeat.
3	**Ridout Q_m^{-2} Critical Tangency**	`SB.75` (two-window $2^{T_r}/3^{n_2}$ tail cancellation identity; `RidoutSubspaceFramework.lean`).	Ridout (1957): canceling 2^{T_m} between u_m and Δ_m lands at exact product exponent Q_m^{-2}, missing $Q_m^{-2-\varepsilon}$.
4	**Multi-Origin Subspace Syzygy**	Root `MultiOriginSubspaceSyzygy.lean` (exact 2D linear dependence of ≥ 3 shifted origins).	Schlickewei (1990): ≥ 3 shifted origins lie in the 2D subspace $\mathrm{span}\{\Delta_m^{(0)}, 3^{T_m}\}$, triggering the degenerate exception.
5	**Pólya–Carlson Capacity $R_\infty R_2 = 1$**	`SB.3`, `SB.117` ($2\times 2, 3\times 3, 4\times 4$ Hankel parity reduction; $H_n^{(4)} = 0$ on 3-block $s = 2$ runs).	Pólya (1921): $G(z) = \sum g_n z^n \in \mathbb{Z}[[z]]$ has meromorphic radius 1 in $\mathbb{C}$ and 1 in $\mathbb{C}_2$, sitting on $R_\infty R_2 = 1$.
6	**Parseval Square-Root Floor & $1 \to 0$ Gap**	`SB.121–SB.122`, `SB.124` ($2^{4W-3} < 3^{3W-2}$), `SB.128–SB.129` (≤ 1 positive element if 0 triples), `SB.135`, `SB.147`.	Plancherel forces $\max_{h \text{ odd}} \lvert\mathcal{S}(h;N)\rvert \geq \sqrt{U_{N-1}}$; $b_0 = 0$ always survives, blocking unsigned L^1 and 3-point methods at $1 \to 0$.
7	**Pigeonhole Excess, abc Product, & Rogue Resets**	`SB.133` ($12k+1 < 17^k \leq \lvert\mathcal{C}_{7k}\rvert$), `SB.134` ($2^{S_0} u_K < u_K R_0$), `SB.150` ($\leq 1$ rogue d per q when $2R \leq 3^4$).	Generic coprime squarefree $u_K R_0 \asymp C^{1.3691} > C$ exceeds abc, and bridging $\langle 3\rangle$-cosets to short bands faces variable-d resets.

34. Verification Instructions

Because `SemigroupBreakoutTheorems.lean` imports the root synthesis module `SubCeilingKneadingAutomatonAndGrandSynthesis54` while remaining cordoned off from `lakefile.toml`, first build the root library once from the `mahler_demand_a` directory and then verify the 154-theorem breakout module with warnings treated as errors:

```
# 1. Build the 226-file root library (generates required .olean dependencies):
lake build

# 2. Verify all 154 Semigroup Breakout theorems (0 sorrys, 0 warnings):
lake env lean -DwarningAsError=true semigroup_breakout\SemigroupBreakoutTheorems.lean
```

To run the complete 35-script Python computational discovery and verification suite from `mahler_demand_a`:

```
python semigroup_breakout\loop1_product_formula_and_pade.py
python semigroup_breakout\loop1_beukers_pade_and_hankel.py
python semigroup_breakout\loop2_pade_lll_and_divergence.py
python semigroup_breakout\loop3_archimedean_ceiling_and_small_u.py
python semigroup_breakout\loop4_flp_sparsity_and_repulsion.py
python semigroup_breakout\vibe_mathing_wildcards.py
python semigroup_breakout\vibe_mathing_dual_digitsum_tower.py
python semigroup_breakout\vibe_mathing_ideas_1_2_3.py
python semigroup_breakout\sandbox_symmetry_breaking_loops.py
python semigroup_breakout\sandbox_wave2_towers.py
python semigroup_breakout\sandbox_wave3_towers.py
python semigroup_breakout\sandbox_wave4_canyon_view.py
python semigroup_breakout\sandbox_wave5_multi_block_crt.py
python semigroup_breakout\sandbox_wave6_both_bridges.py
python semigroup_breakout\sandbox_wave7_binary_carry_bridge.py
python semigroup_breakout\sandbox_overnight_20_bridge_loops.py
python semigroup_breakout\sandbox_wave8_jk_diagonal_asymptotics.py
python semigroup_breakout\sandbox_wave9_seed_finiteness_and_extrema.py
python semigroup_breakout\sandbox_wave10_lte2_baker_and_finiteness.py
python semigroup_breakout\sandbox_wave11_upgraded_baker_and_return_extinction.py
python semigroup_breakout\sandbox_wave12_global_dominance_and_product_formula.py
python semigroup_breakout\sandbox_wave13_kempner_two_adic_expansion.py
python semigroup_breakout\sandbox_wave14_ceiling_dual_gauge_tower.py
python semigroup_breakout\sandbox_wave15_superblock_and_multigauge.py
python semigroup_breakout\sandbox_wave16_velocity_calculus_and_master_closure.py
python semigroup_breakout\sandbox_wave17_path_a_and_path_b.py
python semigroup_breakout\sandbox_wave18_mod40_and_multibase_tower.py
python semigroup_breakout\sandbox_wave19_superblock_sieve_and_density.py
python semigroup_breakout\sandbox_wave20_universal_q_half_relaxation_and_kempner_dichotomy.py
python semigroup_breakout\sandbox_wave21_all_eight_angles_0_to_7.py
```

```
python semigroup_breakout\sandbox_wave23_fourier_frontier_and_parseval_barrier.py
python semigroup_breakout\sandbox_wave24_paths_a_and_b_deep_exploration.py
python semigroup_breakout\sandbox_wave25_method2_ideas_and_real_gauge_pinch.py
python semigroup_breakout\sandbox_wave26_twisted_operator_and_laplace_partition.py
python semigroup_breakout\sandbox_wave27_capstone_rank_compression_and_barrier7.py
```

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35. Acknowledgments & References

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